Exact Compositional Transfer of Bounded Linear Recovery

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Abstract

Linear recovery under concatenation depends on which intermediate functional an outer equation requires, not only on the least helper count. We give an exact labelled min–sum formula for the first recovery equation leaving its target block and prove associative composition with coefficient witnesses. Within one block, relative generalized Hamming weights of the shortening–puncturing pair give the minimum helper unions in each recovered dimension. Operational recovery admits a stronger transfer theorem: if every outer dual word touching the target block has weight greater than $r + 1$, all inclusion-minimal recovery supports through radius r remain local, without the additive inner-dual restriction needed for equation confinement. This transfers reliability under arbitrary availability laws with the same local marginal, fractional service regions, and integral support allocations. A general quotient-labelled lifting theorem composes prescribed operations in nested code pairs. Support antichains preserve resource alternatives; all nonnegative price queries contain exactly the same information as the availability function. Boundary coordinates give fixed-query complexity bounds, and bilinear worker tasks supply a worked extension. The bounded numerical contextual quotient remains a coarser observation contract. Ergodis implements related compilation, optimization, and verification interfaces; no proof depends on its execution.

1 Introduction

Which recovery information survives concatenation? A scalar locality or confinement threshold does not: the outer code can force an intermediate functional whose cost was discarded when the scalar minimum was taken. The problem is therefore compositional. We seek finite interfaces that compile recovery choices through successive concatenations while preserving the distinctions a specified objective observes.

A linear recovery equation expresses a target combination as a linear combination of helper symbols; its cost is the number of distinct helpers used. It is *confined* when every coefficient lies in the target's own inner block and *nonconfined* when at least one helper lies in another block.

The information loss can be seen before any coding notation. Suppose an inner block can realize two intermediate functionals a and b at respective costs one and two. The scalar minimum is one. If an outer equation forces label b, however, the relevant cost is two; replacing the labelled table $(a \mapsto 1, b \mapsto 2)$ by its minimum has destroyed the answer. The exact theory below makes this schematic example intrinsic and, in Proposition 4.3, realizes it while holding the unlabelled recovery data fixed.

At the coefficient level, exact transfer means more than equality of a numerical parameter. Under the equation-confinement criterion below, every global recovery is uniquely the zero-extension of a recovery inside the target block. Formally, restriction to that block and zero-extension back into the concatenation are inverse bijections on normalized recovery-equation systems, preserving coefficients and exact helper supports. Normalization means that the target coefficients act as the identity on the recovered subspace. Operational queries need only minimal supports. Theorem 4.2 transfers those under the weaker condition $\delta_j(O^{\perp}) > r+1$, where δ_j is the least weight of an outer dual word touching the target block. Unrelated external equations can violate coefficient confinement without creating a new minimal recovery option.

Three projections of the recovery data answer different questions. Relative generalized Hamming weights give the minimum helper union at each recovered dimension. Exact helper-support families retain the overlap data needed for reliability. Labelled prescribed-coset support functions start from the same equations but fix the induced intermediate functional before minimizing, which keeps the labels needed for concatenation. Figure 1 shows how these projections meet; neither the support-family branch nor the labelled-cost branch contains the other.

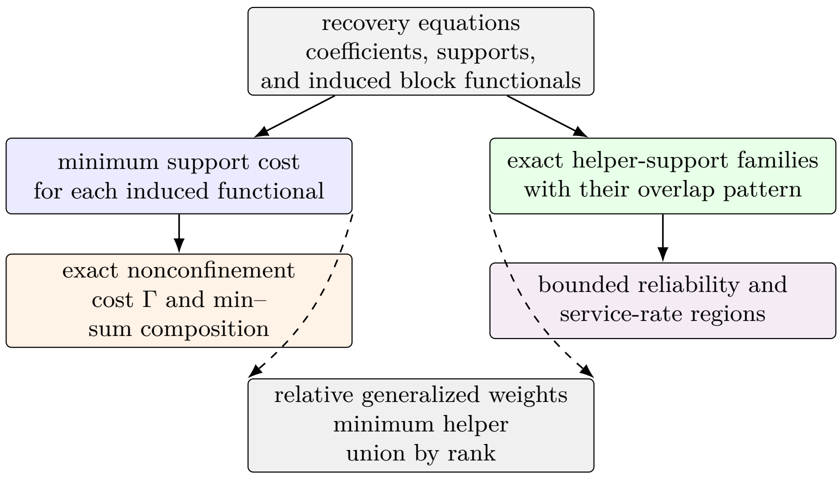

Figure 1: Two projections of coefficient-labelled recovery equations answer different questions. On the left, minimizing within each induced-functional class gives the labelled costs used for exact numerical composition; a solver must also retain a minimizing equation to propagate coefficient witnesses. Exact support families retain the overlaps needed for stochastic and fractional recovery questions. Relative weights are the common minimum-cost projection (dashed arrows), complete for helper-union minima but not for either richer question. Labelled support antichains refine both branches for operational queries: they compose before forgetting labels and retain all nonnegative helper-price responses.

Question	Retained information and exact result
How many helpers recover a t-dimensional subspace inside one block?	The associated nested pair gives an exact relative generalized Hamming weight.
When can a recovery first leave its target block?	Minimum support in each induced-functional class gives an exact two-sector nonconfinement cost.
What must pass to the next concatenation level?	The same labelled costs compose by exact associative min–sum substitution; minimizing lifts propagate a witness.
What is the smallest exact bounded state?	A finite contextual response vector identifies exactly the states indistinguishable to every compatible outer problem; at rank one it has an explicit projective description.
When is every recovered dimension confined through radius r?	The rank-one escape cost suffices: every rank is confined if and only if rank one is confined.
What determines bounded reliability and capacity sharing?	Exact minimal helper supports and their overlaps determine the corresponding block-local statistics; target-touching distance suffices for their transfer.

Table 1: Six questions answered by different projections of the recovery data. The last row needs no equation-confinement hypothesis once minimal supports agree.

Let q be a prime power, let $I \leq \mathbb{F}_q^E$, and partition the coordinates as $E = P \sqcup J$ into targets and helpers. The associated nested code pair is

$$D_P = \mathrm{punct}_J(I^\perp), \qquad K_P = \mathrm{short}_J(I^\perp).$$

Here punct_J restricts a word to the helper coordinates J, whereas short_J first requires the target coordinates P to vanish and then restricts to J. For a full-row-rank encoder $G = (G_P \mid G_J)$, put $W_P = \mathrm{im}\, G_P \cap \mathrm{im}\, G_J$. The exact sequence in Proposition 3.1 identifies $D_P/K_P \cong W_P$, the space of target combinations recoverable from the helpers. We assume $0 < \dim I < |E|$, so both the represented message space and $I^\perp$ are nonzero. We use the convention $d(0) = \infty$ for the zero code. Here t is the dimension of a recovered subspace of W_P, not the number of erased coordinate positions. Recovering all values on P is the special case $\mathrm{im}\, G_P \subseteq \mathrm{im}\, G_J$ and $t = \dim \mathrm{im}\, G_P$; dependent target columns may make this dimension smaller than $|P|$.

The first layer is classical nested-code support theory in recovery language: the minimum helper union for recovered dimension t is $M_t(D_P, K_P)$, the tth relative generalized Hamming weight of the pair. Theorem 3.3 gives the exact support-preserving correspondence. The principal result begins at the next layer, where the outer code can distinguish functional labels having the same minimum support.

For concatenation, identify the message space of I with a finite extension field $L/\mathbb{F}_q$ and retain the chosen encoder $L \to I$. We call this encoder together with its image a *represented inner code*; the representation matters because it assigns field functionals to coefficient fibres. Section 4 groups recovery equations by the functional induced on each block and minimizes support within every such class. Its exact two-sector value is denoted by $\Gamma_{j,T}(O, I)$; schematically,

$$\Gamma_{j,T} = \min\{\text{zero-label sector},\ \text{least compatible nonzero-label block sum}\}.$$

The zero-label sector is one inner recovery plus a least inner-dual perturbation in another block; a nonzero label prescribes one coefficient class in every selected block. Put $\Gamma_{j,t}(O, I) = \min\{\Gamma_{j,T}(O, I) : T \leq W_P,\ \dim T = t\}$.

Theorem 1.1 (Main characterization: exact finite transfer)**.** *Let $\ell = \dim W_P \geq 1$ and $1 \leq t \leq \ell$. Let I be used as the represented inner code in a concatenation with an L-linear outer code $O \leq L^N$ with $N \geq 2$, where L is the message space of I, and fix a block j onto which O projects nontrivially. The minimum cost of a nonconfined recovery system—one using at least one helper outside block j—in recovered dimension t is $\Gamma_{j,t}(O, I)$. Hence confinement through radius r holds exactly when $r < \Gamma_{j,t}(O, I)$.*

The content is the correspondence behind the formula: every global recovery system determines exactly one compatible labelled family in one of its two terms, and every such family lifts to a global recovery with the displayed sum of block supports.

Outer-distance collapse. Sufficient outer dual distance rules out every competing nonzero-label sector, leaving the relative-weight cost plus one inner-dual perturbation. The exact characterization gives

$$d(O^{\perp}) \geq M_t(D_P, K_P) + d(I^{\perp}) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^{\perp}).$$

For a fixed radius r, the weaker condition $d(O^{\perp}) > r + 1$ reduces the exact criterion to

$$r < M_t(D_P, K_P) + d(I^{\perp}).$$

When this outer-distance condition and the displayed inequality both hold, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. Consequently, for any outer family with dual distance tending to infinity, the same statement holds for all sufficiently long concatenations at every fixed r.

The quotient-lifting law in Theorem 3.5 places recovery in a general nested-pair framework. Its support-antichain refinement retains resource choices, and all nonnegative helper prices determine the same minimal family as availability queries (Proposition 6.2). Scalar labelled costs give the specialized closure law in Theorem 4.4: the labelled prescribed-coset functions entering $\Gamma_{j,T}$ compose associatively by exact min–sum substitution. Storing an argmin in each attained fibre also propagates one coefficient-level witness; the numerical state itself contains only the fibre minima.

The closed labelled state is generally redundant. At rank one, Theorem 4.6 quotients it by indistinguishability under every compatible finite outer context and identifies the minimal numerical response exactly: the zero-sector cost and the cheapest response to each projective functional line an outer code can probe. In optimization language, this is a Myhill–Nerode-type minimization theorem: two rank-one states are equivalent exactly when no compatible outer problem distinguishes their optimal costs, and the resulting coarsest quotient is a congruence for further concatenation. The higher-rank statement is finite but less explicit. Through helper radius r, Theorem 4.7 shows that contexts of length at most $\max\{2, r + 1\}$ and functional-dual dimension at most $\min\{t, r\}$ detect every distinction. Their finite response vector therefore gives the coarsest exact bounded numerical quotient at rank t, and the same semantic argument makes it a congruence. Thus the closed labelled state supplies a convenient recursion, whereas contextual equivalence removes exactly the numerical information that no compatible outer problem can see.

The rank-one consequence is especially simple: a nonconfined recovery on a higher-dimensional target restricts to a nonconfined recovery on a target line without increasing its helper support. Hence confinement through a fixed radius holds at every recoverable rank if and only if it holds at rank one (Corollary 4.9).

The label loss is already visible over $\mathbb{F}_2 \subseteq \mathbb{F}_4$. Two three-column encoders have the same binary image code, dual, abstract nested pair, complete relative-weight hierarchy, dual distance, and unlabelled helper-support family. Their functional labels differ, and one fixed outer functional-dual line gives exact nonconfinement costs one and two. Proposition 4.3 gives the calculation before the closure theorem. Thus every unlabelled and scalar explanation is held fixed; only alignment with the intermediate functional labels changes.

The transfer statements in the last row are block-local: irrelevant cross-block supersets do not change a minimal support, bounded success event, or bounded allocation problem. The one-coordinate specialization and its total-word-weight convention are recorded after the general transfer theorem.

Closest precedents. RGHWs and RDLPs describe single-block support and access thresholds for nested codes [29, 25, 16]; our first identity specializes that theory to the canonical target/helper pair. Unlike recovery-set summaries, the compositional state retains functional labels; its min–sum propagation has a different objective from concatenated decoding, and its contextual quotient is analogous to, but not derived from, weighted automata minimization [17, 2, 5, 35]. Section 4.6 gives the detailed comparisons and citations.

ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) is an separately developed compiler and exact solver for finite algebraic optimization, initially motivated by these reductions. From finite linear recovery data and optional capacity constraints it composes labelled costs, computes confinement thresholds and schedules, and returns replayable coefficient-level witnesses. Section 6 gives correctness and complexity bounds and evaluates representative storage, graph, code-design, and tower applications against matched exact controls. No mathematical result depends on the implementation or its measurements.

For a first pass, Sections 2–4 give the proof spine, and Section 5 shows what its coarser summaries lose. Section 4.6 collects the detailed literature comparison and may be skipped on that pass. The later sections give the executable reduction, consequences, a worked family, and verification.

2 Recovery sets and normalized equations

Let $C \leq \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r. A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \qquad (c \in C).$$

Its exact helper support is $\operatorname{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\leq r)}(C) = \{\operatorname{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \operatorname{wt}(w) \leq r+1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\leq r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r. Its inclusion-minimal members form the minimal recovery-set clutter. Here a clutter is a family of sets none of which contains another. Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^\perp : w_x = 1, \operatorname{wt}(w) \leq r+1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of $\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [42, 45, 20, 33].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \quad \Longleftrightarrow \quad A \text{ contains a member of } \mathcal{H}_x^{(\le r)}(C).$$

Under independent helper survival with probabilities (s_y), the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 5 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery system is a linear family whose image is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. We view its blocks as maps $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$. If $T \le \operatorname{im} G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \qquad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P\alpha = \mathrm{id}_T, \qquad G_J\beta = -\,\mathrm{id}_T\,.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\mathrm{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\,\mathrm{supp}(\beta(T))|$. We call a target subspace helper- recoverable when it lies in

$$W_P = \operatorname{im} G_P \cap \operatorname{im} G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

3 Puncturing, shortening, and relative recovery costs

Continue with $G = (G_P \mid G_J)$, viewed as $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$, and set

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

$$K_P = \ker G_J, \qquad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair)**.** *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \tag{1}$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P. Its image is

$$G_J(D_P) = \operatorname{im} G_J \cap U_P = W_P,$$

so the restricted map is onto W_P. □

The first isomorphism theorem applied to the restricted map gives $D_P/K_P \cong W_P$.

Proposition 3.2 (Canonical puncturing–shortening description)**.** *With puncturing and shortening taken onto the helper set J,*

$$D_P = \mathrm{punct}_J(I^\perp), \qquad K_P = \mathrm{short}_J(I^\perp). \tag{2}$$

Thus punct_J *restricts to J, while* short_J *first requires the coordinates on P to vanish and then restricts to J. These operations are dual to one another. Taking duals inside $\mathbb{F}_q^J$ gives*

$$D_P^\perp = \mathrm{short}_J(I), \qquad K_P^\perp = \mathrm{punct}_J(I). \tag{3}$$

Consequently the nested pair depends only on the code I and the split $E = P \sqcup J$; changing the row basis of a generator matrix leaves it unchanged.

Proof. A vector $a \in \mathbb{F}_q^J$ lies in $\mathrm{punct}_J(I^\perp)$ exactly when some $p \in \mathbb{F}_q^P$ satisfies $G_P p + G_J a = 0$, equivalently $G_J a \in \operatorname{im} G_P$, which is $a \in D_P$. It lies in $\mathrm{short}_J(I^\perp)$ exactly when $(0, a) \in I^\perp$, equivalently $G_J a = 0$, which is $a \in K_P$. The dual identities follow from the standard puncturing–shortening duality. □

Thus a monomial relabeling of helpers gives the corresponding monomially equivalent pair, while a row-basis change does nothing to the pair itself.

For a subspace $A \leq \mathbb{F}_q^J$, write $\operatorname{supp} A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the tth relative generalized Hamming weight is

$$M_t(D, K) = \min\{|\operatorname{supp} L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t. Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [29]; standard treatments also cover ordinary Wei duality and computational forms [16, 41]. For nested code pairs in linear secret sharing, RDLPs and RGHWs already give exact information and reconstruction thresholds [25].

The following result is the standard nested-code support invariant applied to the canonical puncturing–shortening pair, with the normalization and helper support correspondence stated in recovery language.

Theorem 3.3 (Relative weights are exact recovery costs)**.** *Let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The minimum helper-union size among recovery systems for all helper-recoverable t-dimensional subspaces of target linear combinations, denoted $\mu_t(P)$, is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, after forgetting the target normalization, normalized recovery systems of recovered dimension t are in support-preserving correspondence with subspaces

$$L \leq D_P, \qquad \dim L = t, \qquad L \cap K_P = 0,$$

while normalization on the chosen recovered subspace $T \leq W_P$ is a linear section of G_P over T.

Proof. Let a normalized system recover a t-dimensional subspace $T \le W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\operatorname{id}_T$, the map β is injective, $L \le D_P$, and $L \cap K_P = 0$. Moreover $G_J L = T$, and the union of helpers in the system is exactly $\operatorname{supp} L$.

Conversely, let $L \le D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t-dimensional subspace $T = G_J L \le W_P$. Choose a linear section $\alpha : T \to \mathbb{F}_q^P$ with $G_P\alpha = \operatorname{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\operatorname{supp} L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \le D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L'. Then $\dim L = t$, $L \cap K_P = 0$, and $\operatorname{supp} L \subseteq \operatorname{supp} L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. □

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \bigl(\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)\bigr),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H.

Proposition 3.4 (Recovery interpretation of the relative profile)**.** *For $0 \le s \le |J|$,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \dim\bigl(W_P \cap \operatorname{span}(G_H)\bigr).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \ge t\}.$$

Thus K_s is the maximum number of independent target linear combinations recoverable from a fixed budget of s helpers.

Proof. Fix H. Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \operatorname{span}(G_H)$. Rank–nullity gives the first equality after maximizing over H. The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.3. □

Standard shortening–puncturing duality gives the complementary failure interpretation: the relative weights of $D_P^\perp \subseteq K_P^\perp$ are the minimum numbers of helper failures that leave prescribed dimensions of W_P ambiguous [29, 16].

The strict inequalities

$$1 \le M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that each additional independent target combination requires a strictly larger helper union. These minima solve the single-block problem but do not yet provide recursive state: the next section restores the functional labels needed for exact concatenation. Only when an outer-distance condition removes every nonzero label does the answer collapse to $M_t(D_P, K_P) + d(I^\perp)$.

3.1 Prescribed quotient operations

The pair construction admits a useful general form. For finite linear codes $K \subseteq D \leq \mathbb{F}_q^E$, let $\pi : D \to D/K$ be the quotient map. Given a finite $\mathbb{F}_q$-space T and a linear map $B : T \to D/K$, put

$$\kappa_T^{D/K}(B) = \min_{\substack{Y:T\to D \text{ linear} \\ \pi Y = B}} |\operatorname{supp} Y(T)|.$$

Thus a quotient label specifies an operation and a lift specifies its physical coordinates. Under $D_P/K_P \cong W_P$, the inclusion of a target subspace T has cost $\rho_T(I)$: its helper lift has image under G_J equal to the requested target, and negating the helper coefficients gives the normalized dual equations. More generally, minimizing $\kappa_T^{D/K}(B)$ over injective B with $\dim T = t$ gives $M_t(D, K)$. Indeed an injective quotient image forces the lift image to be disjoint from K, and every t-dimensional subspace disjoint from K supplies such a map after choosing a basis.

Theorem 3.5 (Composition of quotient-labelled lifts)**.** *Let E_i be disjoint, let $K_i \subseteq D_i \leq \mathbb{F}_q^{E_i}$, and write $\pi_i : D_i \to L_i = D_i/K_i$. For a nested pair $R_0 \subseteq R_1 \leq \bigoplus_i L_i$, take inverse images inside $\bigoplus_i D_i$:*

$$\pi = \bigoplus_i \pi_i, \qquad D = \pi^{-1}(R_1), \qquad K = \pi^{-1}(R_0).$$

Then $D/K \cong R_1/R_0$. Writing $\bar{\pi} : R_1 \to R_1/R_0$, every linear $B : T \to R_1/R_0$ satisfies

$$\kappa_T^{D/K}(B) = \min_{\substack{Z:T\to R_1 \text{ linear} \\ \bar{\pi} Z = B}} \sum_i \kappa_T^{D_i/K_i}(\operatorname{pr}_i Z). \tag{4}$$

Minimizing lifts compose to a witness. Repeated substitution is associative.

Proof. The surjection $D \to R_1/R_0$ induced by π has kernel K, giving the asserted isomorphism. A lift $Y : T \to D$ determines $Z = \pi Y$ with $\bar{\pi} Z = B$. For fixed Z, its block maps Y_i are precisely the independent lifts of $\operatorname{pr}_i Z$. Their supports are disjoint, so their minimum costs add. All quotient maps admit linear sections; hence these minima are attained and their block witnesses assemble to a lift of B. This proves both inequalities in the formula. At further levels, either order of substitution minimizes over the same compatible intermediate maps and the same sum of leaf costs. ☐

No common extension field is needed here. For heterogeneous injective encoders $e_i : V_i \hookrightarrow \mathbb{F}_q^{E_i}$ and a base-field-linear outer relation $O \leq \bigoplus_i V_i$, the interface is $e_i^* : \mathbb{F}_q^{E_i} \to V_i^*$ and compatibility is membership in $\operatorname{ann}(O)$. Orthogonality to all encoded outer words proves this directly. Target-normalized fibres retain their target constraints and exclude target coordinates from the cost. To identify a prescribed target with the full alphabet V_j, the projection $O \to V_j$ must be surjective; a nonzero projection alone is insufficient. With nonzero T and at least two blocks, the zero-functional nonconfinement contribution is $\rho_{j,T} + \min_{i\neq j} d(I_i^\perp)$, by adjoining a minimum dual word in the cheapest external block. Extension-field scalar and projective actions used below are additional structure of that specialization.

Available quotient directions. For a coordinate set $A \subseteq E$, the quotient directions realizable entirely on A form

$$L(A) = \pi(D \cap \mathbb{F}_q^A), \qquad \dim L(A) = \dim(D \cap \mathbb{F}_q^A) - \dim(K \cap \mathbb{F}_q^A).$$

Here $\mathbb{F}_q^A$ is embedded by zero-extension. The formula is rank–nullity for the restricted quotient map. A prescribed B has a lift on A exactly when $B(T) \subseteq L(A)$, since the restricted surjection admits a linear section. This retains the directions exposed on A, as well as their maximum possible rank; minimizing ordinary words of D would lose the requirement that the quotient label be nonzero.

4 Exact confinement under concatenation

The argument has two stages. First, every concatenated recovery system is classified by its outer functional label. The zero label gives a local recovery system plus an inner-dual perturbation; each nonzero label prescribes a coset in every active block. Minimizing these two sectors gives the exact finite nonconfinement cost. Second, the full label-indexed cost functions, rather than their global scalar minima, are passed to the next concatenation level by an associative min–sum law.

Represented inner code. Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \leq \mathbb{F}_q^E$ be an $\mathbb{F}_q$-linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. Whenever the outer alphabet L occurs below, I denotes the represented inner code (I, ι), not only its image subspace. In particular, $O \circ I$, Φ_I, $\lambda_{I,T}$, $\mu_{I,P,T}$, and $\Gamma_{j,T}(O, I)$ depend on ι. The earlier pair $K_P \subseteq D_P$ depends only on the image code and target/helper split. Proposition 4.3 shows that this distinction cannot be suppressed under concatenation.

For an L-linear outer code $O \leq L^N$, write

$$O \circ I = \{(\iota(a_1), \dots, \iota(a_N)) : (a_1, \dots, a_N) \in O\}.$$

Each block has a label map

$$\underbrace{\mathbb{F}_q^E}_{\text{inner coefficients}} \xrightarrow{\Phi_I} \underbrace{L^*}_{\text{functional label}}, \qquad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. Two coefficient vectors receive the same label precisely when they differ by an inner dual word, because $\ker \Phi_I = I^\perp$. The prescribed-coset costs below record the cheapest coefficient support in each fibre of this map.

Outer compatibility. A block vector $y = (y_1, \dots, y_N)$ belongs to $(O \circ I)^\perp$ precisely when

$$(\Phi_I(y_1), \dots, \Phi_I(y_N)) \in \mathrm{FD}(O), \tag{5}$$

where

$$\mathrm{FD}(O) = \{(\beta_j) \in (L^*)^N : \textstyle\sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Write $\beta_j(a) = \mathrm{Tr}_{L/\mathbb{F}_q}(b_j a)$. The functional-dual condition becomes

$$\mathrm{Tr}_{L/\mathbb{F}_q}\left(\sum_j b_j a_j\right) = 0 \qquad \text{for every } (a_j) \in O.$$

Because O is L-linear, the same holds after multiplying (a_j) by every $\lambda \in L$. Nondegeneracy of the trace pairing then forces $\sum_j b_j a_j = 0$. Hence $(b_j) \in O^\perp$, and the converse is immediate. This identification preserves block support, so the support distance of $\mathrm{FD}(O)$ is $d(O^\perp)$.

Confinement and target normalization. Fix a target block and a nonzero target-message subspace $T \leq U_P$. A system is *confined* if every one of its equations is supported in that target block. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code, with value ∞ when no such system exists. Thus $\rho_T(I) < \infty$ exactly when $T \leq W_P$. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P\alpha = \mathrm{id}_T$ under the fixed inner-message identification. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^\perp) > 1$ guarantees surjectivity: an L-linear coordinate image is either 0 or L, and a zero image would put a weight-one word in $O^\perp$.

For an $\mathbb{F}_q$-linear map $B : T \to L^*$, define

$$\lambda_{I,T}(B) = \min_{\substack{Y:T\to\mathbb{F}_q^E \text{ linear}\\ \Phi_I Y = B}} |\operatorname{supp} Y(T)|. \tag{6}$$

For the target block, define

$$\mu_{I,P,T}(B) = \min_{\substack{\alpha:T\to\mathbb{F}_q^P,\ \beta:T\to\mathbb{F}_q^J \text{ linear}\\ G_P\alpha=\mathrm{id}_T\\ \Phi_I(\alpha,\beta)=B}} |\operatorname{supp}\beta(T)|. \tag{7}$$

An empty minimum is ∞. After choosing a basis of T, (6) is the minimum union support of representatives of prescribed cosets of $I^\perp$. This is the fixed-instance quantity underlying generalized coset weights and generalized covering radii [46, 10]. The target version retains the normalization required by the recovery problem. In particular,

$$\mu_{I,P,T}(0) = \rho_T(I). \tag{8}$$

For each equation parameter $t \in T$, the value $B(t)$ is exactly the outer functional label induced by that equation in the block.

Let

$$\mathcal{B}_T(O) = \operatorname{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

For $B \in \mathcal{B}_T(O)$, write $B = (B_h)_{h=1}^N$ with $B_h : T \to L^*$. Assume $N \geq 2$, fix a target block j whose coordinate projection $O \to L$ is nonzero, hence surjective, and put

$$\Gamma_{j,T}(O,I) = \min\left\{ \mu_{I,P,T}(0) + d(I^\perp), \; \min_{0\neq B\in\mathcal{B}_T(O)} \left(\mu_{I,P,T}(B_j) + \sum_{h\neq j}\lambda_{I,T}(B_h)\right)\right\}. \tag{9}$$

Theorem 4.1 (Exact nonconfinement cost from labelled coset supports)**.** *Under the hypotheses above, $\Gamma_{j,T}(O,I)$ is the least helper-union size of a nonconfined normalized recovery system for T in $O \circ I$, with value ∞ if no such system exists. Thus every cost-$\leq r$ system for T is confined if and only if*

$$r < \Gamma_{j,T}(O,I). \tag{10}$$

For $1 \leq t \leq \ell = \dim W_P$, set

$$\Gamma_{j,t}(O,I) = \min_{\substack{T\leq W_P\\ \dim T = t}} \Gamma_{j,T}(O,I). \tag{11}$$

Then every cost-$\leq r$ system for every helper-recoverable t-dimensional target subspace is confined if and only if $r < \Gamma_{j,t}(O,I)$. Below either threshold, restriction and zero-extension are inverse bijections on normalized systems and preserve exact helper supports.

Proof. Write a recovery system as an $\mathbb{F}_q$-linear map $Y : T \to (O \circ I)^\perp$ and let Y_h be its block maps. By (5),

$$B_h = \Phi_I Y_h$$

defines a linear map $B : T \to \mathrm{FD}(O)$. Conversely, compatible linear lifts of the block maps of any such B give a map into the concatenated dual. At the target block, compatibility with the chosen target space is exactly the constraint in (7).

Suppose first that $B \neq 0$. No nonzero element of $\mathrm{FD}(O)$ is supported only at j: testing such an element against outer words whose jth coordinate ranges over L would force its jth functional to vanish. Hence some block map B_h with $h \neq j$ is nonzero, and every lift is nonconfined. Distinct inner blocks have disjoint helper coordinates, so the minimum helper union for this fixed B is

$$\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h).$$

The block lifts can be chosen independently, so this lower bound is attained.

If $B = 0$, every block map takes values in $I^\perp = \ker \Phi_I$. The target block costs $\rho_T(I) = \mu_{I,P,T}(0)$. Nonconfinement requires a nonzero map $T \to I^\perp$ in another block. Its image has union support at least $d(I^\perp)$, and equality is attained by $u \mapsto f(u)v$, where $0 \neq f \in T^*$ and v is a minimum-weight word of $I^\perp$.

The zero and nonzero functional sectors are exhaustive, proving (9) and (10). Minimizing over t-dimensional T proves the ranked statement. Below the first nonconfined cost, every global system has zero coefficient maps in all blocks outside j. Restriction therefore preserves its target normalization and helper support. Conversely, zero-extending an inner dual system remains orthogonal to every concatenated word, so the two operations are inverse with coefficients and supports unchanged. □

A complete small instance. Let $I \leq \mathbb{F}_2^3$ have generator columns $G = (e_1 \mid e_2, e_1 + e_2)$, with the first coordinate targeted. Then $K_P = 0$, $D_P = \langle(1,1)\rangle$, and $M_1(D_P, K_P) = 2$: the target is recovered from exactly the two helpers. Moreover $I^\perp = \langle(1,1,1)\rangle$, so the zero-functional nonconfinement cost is $2 + 3 = 5$.

Identify the two-dimensional message space with $L = \mathbb{F}_4$ and concatenate with the length-six single-parity-check code over L. Its functional dual is the repetition line. Every nonzero label is active in all five blocks outside the target block. An active block has a nonzero functional label, so every lift has nonempty support and costs at least one helper. Thus its nonzero-functional sector also costs at least five, so $\Gamma_{j,1} = 5$. Every recovery system of cost at most four is therefore confined, while a cost-five witness is obtained by adjoining the inner-dual word $(1, 1, 1)$ in another block.

The later scheduler uses the same support data but asks a different question. Two demands in distinct blocks fit unit helper capacities, whereas two demands in one block compete for the same helper pair. Thus this instance will serve as the model case for both composition and scheduling without identifying the two optimizations.

4.1 Minimal supports and operational transfer

An external equation can violate coefficient confinement without supplying a new recovery option. To separate these questions, let $\mathcal{M}^{(\leq r)}_{j,T}(O \circ I)$ be the inclusion-minimal helper supports recovering T with at most r helpers. A set recovers T when some normalized system uses only that set. Define the target-touching distance

$$\delta_j(O^\perp) = \min\{\mathrm{wt}(u) : u \in O^\perp,\ u_j \neq 0\},$$

with empty minimum ∞.

Theorem 4.2 (Minimal-support confinement)**.** *Let $O \le L^N$ be L-linear, let its projection onto block j be surjective, and let $0 \ne T \le U_P$. If $\delta_j(O^\perp) > r+1$, then*

$$\mathcal{M}^{(\le r)}_{j,T}(O \circ I) = \{\{j\} \times H : H \in \mathcal{M}^{(\le r)}_{T}(I)\}.$$

No inequality involving $\rho_T(I) + d(I^\perp)$ is required. Consequently bounded recovery success has the same probability for every global helper-availability law with the prescribed target-block marginal.

Proof. Let Y be a normalized system with helper support H, $|H| \le r$, and put $B_h = \Phi_I Y_h$. If $B_j \ne 0$, choose $v \in T$ with $B_j(v) \ne 0$. The trace identification sends $B(v)$ to an outer dual word touching j. Its other nonzero blocks each contain a helper in H, so its weight is at most $r+1$, a contradiction. Thus $B_j = 0$, and the target-block portion of Y is an inner-dual system with the same target normalization. Deleting all external coefficients preserves recovery and only decreases support. Every global minimal support is therefore local. Conversely, a proper globally recovering subset of a local minimal support would, by this same restriction argument, contradict local minimality. Zero-extension supplies the other inclusion. For each available set, success means containing one of these minimal supports; the two success indicators therefore agree pointwise after restriction to the target block. Taking expectations proves the probability assertion without an independence assumption. □

The relevant first new alternative can also be defined exactly. Let $\varepsilon_{j,T}$ be the least size of a globally recovering helper set H whose portion in block j does not recover T locally. Equivalently, it is the least size of a global minimal support using an external helper: minimize inside H for one direction; a local recovery inside a minimal support contradicts minimality for the other. The theorem gives $\varepsilon_{j,T} \ge \delta_j(O^\perp) - 1$ when the latter is finite, and $\varepsilon_{j,T} = \infty$ when $\delta_j(O^\perp) = \infty$. Its value cannot be obtained simply by deleting the zero sector from $\Gamma_{j,T}$: a nonzero-labelled lift can also contain a local recovery. For $O = L^N$, every minimal recovery support is local at every radius, although adjoining an external inner-dual word gives the finite equation threshold $\rho_T(I) + d(I^\perp)$ whenever T is locally recoverable.

4.2 Repeated concatenation

The ordinary prescribed-coset costs retain the intermediate labels needed at the next concatenation level. Target-normalized composition also retains the part of each intermediate label already supplied by target coordinates. In schematic form,

$$\text{cost of output label} = \min_{\text{compatible intermediate labels}} \sum_{\text{inner blocks}} \text{inner realization cost}.$$

The eliminated variables are the intermediate labels shared by adjacent levels. The resulting formulas are instances of variable elimination over the min-sum semiring [2]. This is also the algebraic form of passing symbol-indexed inner costs to an outer decoder, as in soft and generalized-concatenated decoding [19, 5].

The zero-functional nonconfinement cost alone is not compositional. The next separation holds the underlying binary code and all its unlabelled recovery data fixed.

Proposition 4.3 (Unlabelled data do not determine finite composition)**.** *Over $\mathbb{F}_2 \subseteq \mathbb{F}_4$, there are two represented inner encoders with the same image code, dual code, target/helper nested pair,* *complete relative-weight hierarchy, dual distance, unlabelled normalized helper-support family, and zero-functional nonconfinement costs for every recoverable target subspace, but one fixed outer represented code gives different exact nonconfinement costs.*

Proof. Write the nonzero elements of $\mathbb{F}_4$ as $1, \omega, \omega^2 = 1 + \omega$ and take generator columns

$$I_1 : (1, 1, \omega), \qquad I_2 : (1, 1, \omega^2),$$

with the first coordinate targeted. Relative to the binary basis $(1, \omega)$ of $\mathbb{F}_4$, each displayed field element is a two-dimensional binary generator column. Both generator matrices have binary row space

$$\{(x, x, y) : x, y \in \mathbb{F}_2\}$$

and dual $\langle(1, 1, 0)\rangle$. Thus both have dual distance 2, $K_P = 0 \subseteq D_P = \langle(1, 0)\rangle$, $M_1 = 1$, and the same unique minimal helper support. Their recoverable target space is one-dimensional, so their complete zero-functional profile is the common value $1 + 2 = 3$.

In the label order $(1, \omega, \omega^2)$, the ordinary coset costs are $(1, 1, 2)$ for I_1 and $(1, 2, 1)$ for I_2; the normalized target costs are respectively $(0, 2, 1)$ and $(0, 1, 2)$. Take the fixed outer code whose functional dual is the line spanned by $(1, \omega)$. Its nonzero multiples have target label s and external label $s\omega$. The complete calculation is

s	s	$s\omega$	$\mu_1(s)$	$\lambda_1(s\omega)$	total	$\mu_2(s)$	$\lambda_2(s\omega)$	total
1	1	ω	0	1	1	0	2	2
ω	ω	ω^2	2	2	4	1	1	2
ω^2	ω^2	1	1	1	2	2	1	3

Equation (9) therefore gives exact costs 1 and 2. Only the alignment of costs with the intermediate functional labels has changed. ☐

Let $\mathbb{F}_q \subseteq L \subseteq M$ be a tower of finite fields. Let B be an $\mathbb{F}_q$-linear represented code with message space L, and let A be an L-linear represented code with message space M. The composite $A \circ B$ replaces each coordinate of an A-word by its represented B-block. After identifying a functional with its unique trace coefficient, write their dual restriction maps as

$$\phi_B : \mathbb{F}_q^{E_B} \longrightarrow L, \qquad \phi_A : L^{E_A} \longrightarrow M.$$

For an $\mathbb{F}_q$-space T and $b : T \to L$, put

$$\Lambda_{B,T}(b) = \min_{\phi_B Y = b} |\operatorname{supp} Y(T)|.$$

Define $\Lambda_{A,T}$ and $\Lambda_{A \circ B,T}$ similarly. If $J \subseteq E_B$, let $\Lambda_{B,J,T}$ denote the same minimum with Y supported on J and ϕ_B restricted to those coordinates. Thus ϕ is the trace-coordinate form of Φ, and Λ is the corresponding ordinary prescribed-coset cost λ. The change of letters only distinguishes the two levels in the following substitution.

For the recovery-specific form, suppose that a target set in $A \circ B$ meets block e in $P_e \subseteq E_B$. Put $J_e = E_B \setminus P_e$ and $U_{P_e} = \operatorname{im} \phi_{B,P_e}$. If $T \leq M$, let $\iota_T : T \hookrightarrow M$ be the inclusion. The variables below have the following roles:

U_e	intermediate contribution supplied by target coordinates in block e
X_e	total intermediate functional label required in block e
$X_e - U_e$	label that the helper coordinates in block e must realize.

Theorem 4.4 (Min–sum closure of prescribed-coset support costs)**.** *For every $\mathbb{F}_q$-linear map $c : T \to M$, the ordinary prescribed-coset cost is*

$$\Lambda_{A\circ B,T}(c) = \min_{X:T\to L^{E_A}\ \phi_A X=c} \sum_{e\in E_A} \Lambda_{B,T}(X_e). \tag{12}$$

For $T \le M$ and the target split above, the target-normalized recovery cost is

$$\mu_{A\circ B,P,T}(c) = \min_{\substack{U,X:T\to L^{E_A}\\ U_e(T)\subseteq U_{P_e}\ (e\in E_A)\\ \phi_A U=\iota_T,\ \phi_A X=c}} \sum_{e\in E_A} \Lambda_{B,J_e,T}(X_e - U_e). \tag{13}$$

Both formulas iterate associatively through a finite tower: either parenthesization minimizes over the same intermediate labels and target contributions and sums the same leaf costs. Storing minimizing target and helper lifts propagates one coefficient-level witness; retaining every witness requires the full lift sets rather than only the numerical minima.

Proof. A leaf-level lift $Y : T \to \mathbb{F}_q^{E_A\times E_B}$ determines intermediate maps $X_e = \phi_B Y_e$. Its support is the disjoint union of its supports in the inner blocks. At fixed X, the block lifts are independent and attain their minima separately, which proves (12).

For the target-normalized formula, lift U_e to an actual target map $a_e : T \to \mathbb{F}_q^{P_e}$. The condition $\phi_A U = \iota_T$ is exactly target normalization. Once X_e is fixed, the helper contribution must lift $X_e - U_e$ through ϕ_{B,J_e}, independently in each block. This proves (13); storing the displayed lifts gives the stated witness semantics. At three or more levels, either parenthesization eliminates the same intermediate arrays and target contributions, proving associativity. □

Sharp scalar envelopes. The full labelled function gives simple bounds when only its smallest and largest nonzero values are retained. When $T \neq 0$, set

$$\delta_{B,T} = \min_{0\neq b:T\to L} \Lambda_{B,T}(b), \qquad R_{B,T} = \max_{0\neq b:T\to L} \Lambda_{B,T}(b).$$

For $c \neq 0$, the exact formula has the sharp coarsening

$$\delta_{B,T}\Lambda_{A,T}(c) \le \Lambda_{A\circ B,T}(c) \le R_{B,T}\Lambda_{A,T}(c). \tag{14}$$

Neither constant can be improved without further hypotheses. Indeed, every nonzero intermediate coordinate map costs between $\delta_{B,T}$ and $R_{B,T}$. Minimizing the number of nonzero intermediate coordinates proves the bounds; a one-coordinate outer realization attaining either extremum proves sharpness.

Dual-distance recursion. The same zero/nonzero-label split gives

$$d((A\circ B)^{\perp}) = \min\left\{ d(B^{\perp}),\ \min_{0\neq z\in A^{\perp}} \sum_{e\in E_A} \Lambda_{B,\mathbb{F}_q}(u\mapsto uz_e) \right\}. \tag{15}$$

A word in $(A\circ B)^{\perp}$ either has zero intermediate label, when a minimum word of $B^{\perp}$ in one block is optimal, or has a nonzero label $z \in A^{\perp}$, when the block representatives minimize independently. This proves (15).

Tower consequence. Substituting these recursively computed costs into (9) gives the exact nonconfinement cost through any finite tower, with the zero and nonzero functional sectors kept separate at every level.

The trace identifications in these formulas are compatible with the tower. Write $\iota_A : M \to A \leq L^{E_A}$ for the represented encoder of A. For $v \in T$ and $m \in M$, trace transitivity gives

$$\sum_e \mathrm{Tr}_{L/\mathbb{F}_q}\big(X_e(v)\,\iota_A(m)_e\big) = \mathrm{Tr}_{M/\mathbb{F}_q}\big(\phi_A(X(v))m\big),$$

and nondegeneracy of the trace pairing identifies the composite restriction map with $\phi_A \circ \phi_B^{E_A}$, where $\phi_B^{E_A}$ applies ϕ_B blockwise.

4.3 Contextual equivalence and rank reduction

We first record a normal form for outer functional duals. It will let us replace an arbitrary outer code by the finite interface that a bounded-cost recovery actually touches.

Lemma 4.5 (Pointed column-type normal form)**.** *Let $D = O^{\perp} \leq L^N$ have L-dimension s, and choose an isomorphism $G : L^s \to D$. There are L-linear coordinate covectors $a_h \in (L^s)^{\vee}$ such that*

$$G(x)_h = a_h(x) \qquad (x \in L^s, 1 \leq h \leq N).$$

The outer code projects nontrivially onto the distinguished block j if and only if the external covectors $\{a_h : h \neq j\}$ span $(L^s)^{\vee}$. Conversely, every pointed list $(a_j; (a_h)_{h\neq j})$ with spanning external part realizes such an outer context, uniquely up to change of basis in L^s.

Proof. Take a_h to be the hth coordinate functional of G. The target projection of $O = D^{\perp}$ vanishes exactly when the target coordinate line is contained in D. Under G, this happens exactly when the common kernel of the external covectors contains a vector on which a_j is nonzero. Since G is injective, the common kernel of all a_h is zero, so this is equivalent to the external covectors failing to span. Conversely, the evaluation map defined by a pointed list with spanning external part is injective and has no nonzero target-supported image; taking its image as D gives the required outer code. Changing the basis of L^s changes all covectors by the same linear automorphism and changes nothing else. ☐

For rank-one targets, the exact quotient of the labelled functions observable by outer concatenation has a projective description. Informally, two rank-one inner recovery problems are indistinguishable under every compatible outer concatenation exactly when they agree on the zero-sector cost and on the cheapest response to every projective functional line an outer code can probe. We now encode those probes. Fix a line $T = \langle u \rangle \leq W_P$. For $b \in L$, let $\tau_b : T \to L^*$ be defined by

$$\tau_b(au)(x) = a\,\mathrm{Tr}_{L/\mathbb{F}_q}(bx) \qquad (a \in \mathbb{F}_q,\ x \in L),$$

and put $z_{I,T} = \rho_T(I) + d(I^{\perp})$. If $[b] = [b_1 : \cdots : b_N] \in \mathbf{P}(L^N)$ has a nonzero coordinate outside the target block j, its projective class records an outer functional-dual line. Multiplying by $s \in L^{\times}$ changes the chosen generator of that line but not the outer context, so the probe minimizes over this scalar choice:

$$\Theta_{I,T}([b]) = \min_{s \in L^{\times}} \left(\mu_{I,P,T}(\tau_{sb_j}) + \sum_{h \neq j} \lambda_{I,T}(\tau_{sb_h}) \right). \tag{16}$$

Changing u or the projective representative of $[b]$ does not change this value.

Here an *outer context for the rank-one target T* is a triple (N, j, O) with $N \geq 2$, $O \leq L^N$ an L-linear code, and nonzero projection onto the distinguished block j. The context family ranges over every such N and j, while the inner target/helper split and the identified target line T remain fixed. Thus "rank one" refers to $\dim_{\mathbb{F}_q} T = 1$, not to the dimension of O. Operationally, two inner representations are indistinguishable at T when every compatible outer code assigns them the same exact nonconfinement cost. The theorem below identifies exactly which numerical data determine that response. Two represented inner codes are *contextually equivalent at T* when their values of $\Gamma_{j,T}$ agree in every member of this family. A compatible further concatenation adds outer layers above the same distinguished leaf and does not retarget that leaf.

Theorem 4.6 (Minimal contextual quotient at rank one)**.** *Let $D = O^\perp \leq L^N$ under the trace identification, and suppose the projection of O onto block j is nonzero. Then*

$$\Gamma_{j,T}(O, I) = \min \left\{ z_{I,T}, \min_{\substack{Lb \leq D \\ b \neq 0}} \Theta_{I,T}([b]) \right\}, \tag{17}$$

where Lb is the one-dimensional L-subspace spanned by b, and the inner minimum is empty when $D = 0$.

Consequently, represented inner codes over the same extension field with identified target lines are contextually equivalent at T if and only if their values $z_{I,T}$ agree and the family of truncated profiles

$$[b] \longmapsto \min\{z_{I,T}, \Theta_{I,T}([b])\}$$

agrees for every $N \geq 2$, target block j, and projective tuple with nonzero external part. Thus equality of this response family realizes the exact numerical contextual quotient at rank one: no coarser numerical summary can predict the costs in every outer context. It is enough to test the full outer code and outer codes whose functional dual has L-dimension one. The bounded higher-rank theorem below also shows that, after truncation at any fixed helper radius, only finitely many of these probes are needed. Moreover, contextual equivalence at T is a congruence under the compatible further concatenations defined above.

Proof. A nonzero map from the one-dimensional $\mathbb{F}_q$-space T into D is determined by one nonzero tuple $b \in D$. Partition $D \setminus \{0\}$ into its L-lines in the nonzero-functional sector of (9). Minimization on the line Lb is exactly (16), which proves (17).

It remains to show that every stated probe is an actual outer-code context. The full outer code $O = L^N$ has $D = 0$ and exposes $z_{I,T}$. Given a projective tuple $[b]$ with nonzero external part, take $D = Lb$ and $O = D^\perp$. Its target projection is nonzero: otherwise D would contain the target coordinate line, forcing every external coordinate of b to vanish. This context exposes $\min\{z_{I,T}, \Theta_{I,T}([b])\}$. Equality in every context therefore forces equality of the displayed response family; the first part of the theorem gives the converse.

For the congruence statement, place the same distinguished target leaf in two composites $A \circ I_1$ and $A \circ I_2$. Here compatible means that later layers keep this same target leaf and its induced target image. Any such later context C tests the target-normalized composites $C \circ (A \circ I_i) = (C \circ A) \circ I_i$. The target-normalized composition formula (13) carries the same leaf target image on both parenthesizations, and $C \circ A$ is one of the contexts quantified above. Contextual equivalence of I_1 and I_2 therefore gives equal costs after every such C. Hence contextual equivalence is a congruence for the specified compatible concatenations. ∎

This theorem compares numerical costs only. Full coefficient witnesses still require the minimizing lifts described after Theorem 4.4.

When $\dim_{\mathbb{F}_q} T = t$, the image of any map $B : T \to \mathrm{FD}(O)$ lies in an L-subspace generated by at most t images. A bounded-cost map also uses only boundedly many external labels. Together these observations give a finite small-context theorem. For $c \in \mathbb{N} \cup \{\infty\}$ write $[c]_{r+1} = \min\{c, r+1\}$, with $[\infty]_{r+1} = r + 1$.

Theorem 4.7 (Rank- and radius-bounded outer tests)**.** *Let I be a represented inner code with $0 < \dim I < |E|$, let $0 \neq T \leq W_P$ have $\dim_{\mathbb{F}_q} T = t$, and let $r \geq 0$. Put $D = O^{\perp} \leq L^N$, where $N \geq 2$ and O projects nontrivially onto block j. For L-subspaces $D' \leq D$, set $O_{D'} = (D')^{\perp} \leq L^N$. Then*

$$\Gamma_{j,T}(O, I) = \min_{\substack{D' \leq D \\ \dim_L D' \leq t}} \Gamma_{j,T}(O_{D'}, I). \tag{18}$$

Moreover, for $S \subseteq [N] \setminus \{j\}$ put $D_S = D \cap L^{\{j\} \cup S}$ and $O_S = D_S^{\perp} \leq L^N$, where L^A denotes the vectors supported on A. Then

$$[\Gamma_{j,T}(O, I)]_{r+1} = \min_{\substack{S \subseteq [N] \setminus \{j\} \\ |S| \leq r}} [\Gamma_{j,T}(O_S, I)]_{r+1}. \tag{19}$$

Consequently, if two such inner problems have different truncated responses in some compatible outer context, a separating context exists with length at most $\max\{2, r + 1\}$ and functional-dual dimension at most $\min\{t, r\}$. The target leaf and target image are kept fixed under the identifications used to compare the two problems. The same bounds cover witnesses: every nonconfined normalized coefficient-level system of cost at most r restricts to such a context and zero-extends back to the original system.

Proof. For $B : T \to D$, let $D_B = \operatorname{span}_L B(T)$. The image of an $\mathbb{F}_q$-basis of T generates D_B over L, so $\dim_L D_B \leq t$. Regarding B as a map into D_B changes none of its block labels or costs. Conversely, every map into $D' \leq D$ is a map into D. Minimizing the same cost functional in both directions proves (18); $D' = 0$ supplies the zero-functional sector. No $D' \leq D$ contains a nonzero target-supported vector, so every $O_{D'}$ retains nonzero target projection.

For the truncated identity, a map $B : T \to D$ of cost at most r has at most r nonzero external labels. If S is their set, then $B(T) \leq D_S$, and all block costs are unchanged on passing to O_S. Conversely every map into D_S is a map into D. These two inclusions prove (19); maps of larger cost all truncate to $r + 1$.

Suppose now that two truncated responses differ, and choose a minimizing map for the smaller one. If it is in the zero-functional sector, the full outer code of length two already separates the two inner problems. Otherwise let S be its nonzero external-label set and let $D_B = \operatorname{span}_L B(T)$. Then $1 \leq |S| \leq r$, $\dim_L D_B \leq t$, and the external-coordinate projection is injective on D_B, whence $\dim_L D_B \leq |S|$. Delete every coordinate outside $\{j\} \cup S$ and take the punctured D_B as the new functional dual. The chosen cost is unchanged. Since this is a subcontext of the original one, the other inner problem cannot acquire a smaller value than it had there: every map into the punctured D_B zero-extends to a map into the original D. The resulting context therefore still separates, and it has the claimed length and functional-dual dimension.

Finally, let Y be a nonconfined normalized system of cost at most r, put $B_h = \Phi_I Y_h$, and retain the target block together with the external blocks on which $Y_h \neq 0$. There are at most r of the latter. Replace the functional dual by $\operatorname{span}_L B(T)$ and delete the other blocks. The preceding dimension argument gives the same $\min\{t, r\}$ bound. Keeping activity at the coefficient level, rather than retaining only blocks with $B_h \neq 0$, preserves zero-label inner-dual lifts. Thus Y restricts to the smaller context and zero-extension recovers it exactly. $\square$

Over the finite fields fixed here, choose one representative of every compatible pointed context within the two bounds in the theorem, and list its truncated response. This gives a finite vector $\mathcal{R}_{r,T}(I)$.

Corollary 4.8 (Finite coarsest bounded contextual quotient)**.** *Two represented inner target/helper problems over the same base and extension fields, each with $0 < \dim I < |E|$ and with identified target normalizations on a nonzero space T, have equal values of $\mathcal{R}_{r,T}$ if and only if their truncated nonconfinement costs agree in every compatible finite outer context. Equality of this vector is therefore the coarsest numerical equivalence that determines exact responses through helper radius r. It is a congruence under compatible further concatenation above the same target leaf and induced target image, using the trace-transitive identifications of Theorem 4.4.*

For exact, untruncated comparison, put $z_{I,T} = \rho_T(I) + d(I^{\perp})$. This is finite for every nonzero helper-recoverable T. Two such inner problems are contextually equivalent in all finite outer contexts if and only if their z values agree, say at z, and their exact responses agree on contexts of length at most $\max\{2, z\}$ and functional-dual dimension at most $\min\{t, z-1\}$.

Proof. The theorem says that any unequal bounded responses have a representative among the listed contexts; the converse is immediate. Hence equality of the finite response vector is precisely observational equivalence, and any numerical summary valid in all contexts must refine it. A compatible later context applied to $A \circ I$ is, by associativity and trace transitivity, a context of the form $(C \circ A) \circ I$. Observational equivalence therefore persists under further composition.

The full outer code of length two has response $z_{I,T}$ and separates unequal z values. When both values equal z, every exact response is at most z. Apply the theorem with $r = z - 1$: equality after truncation at z is then exact equality, and the stated finite bounds follow. □

The simultaneous confinement criterion across all recoverable ranks has a single bottleneck.

Corollary 4.9 (Rank-one criterion for complete bounded transfer)**.** *For every target block j with nonzero outer projection,*

$$\min_{0 \neq T \leq W_P} \Gamma_{j,T}(O, I) = \Gamma_{j,1}(O, I). \tag{20}$$

For every integer $r \geq 0$, the following are equivalent:

1. *every cost-$\leq r$ system on every helper-recoverable target line is confined;*
2. *every cost-$\leq r$ system on every nonzero helper-recoverable target subspace is confined; and*
3. *$r < \Gamma_{j,1}(O, I)$.*

Under these conditions, restriction and zero-extension are inverse bijections simultaneously for every nonzero $T \leq W_P$, preserving every coefficient and exact helper support through radius r.

Consequently, any statistic defined solely from these bounded systems and invariant under this bijection is preserved. In particular, the bounded reliability laws and the capacity and scheduling regions defined later from their coefficients or supports agree. Targetwise and rankwise minimum costs also agree after truncation $c \mapsto \min\{c, r+1\}$ (with $\min\{\infty, r+1\} = r+1$). If every relevant inner minimum is at most r, then the untruncated minima agree as well.

Proof. If a system on $0 \neq T \leq W_P$ is nonconfined, some external block map is nonzero on a vector $u \in T$. Restriction to $\langle u \rangle$ remains nonconfined and cannot enlarge its helper union. This proves the displayed equality, and Theorem 4.1 gives the three equivalent conditions. When they hold, every bounded system is confined, so restriction deletes only zero external blocks and zero-extension is its inverse. The coefficient, support, reliability, capacity, and scheduling assertions follow from this bijection. For a minimum helper-union cost, either the inner minimum is at most r and the bijection gives equality, or both inner and concatenated minima exceed r; this is precisely equality after truncation at $r+1$. ☐

4.4 Outer-distance collapse and bounded transfer

The exact optimization also recovers the outer-distance collapse without choosing a radius. Indeed, every nonzero functional sector costs at least $d(O^\perp)-1$: its label has at least $d(O^\perp)$ active blocks, the target block may contribute zero helpers, and every other active block contributes at least one. The zero-functional sector attains $M_t(D_P,K_P)+d(I^\perp)$ after minimization over T. Consequently,

$$d(O^\perp) \geq M_t(D_P,K_P)+d(I^\perp)+1 \quad \Longrightarrow \quad \Gamma_{j,t}(O,I) = M_t(D_P,K_P)+d(I^\perp). \tag{21}$$

Theorem 4.10 (Confinement for a fixed target subspace)**.** *Let r be a nonnegative integer and let $O \leq L^N$ be an L-linear outer code with $N \geq 2$ and $d(O^\perp) > r+1$, and assume $T \leq W_P$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^\perp). \tag{22}$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r.

Proof. If $0 \neq B \in \mathcal{B}_T(O)$, choose $u \in T$ with $B(u) \neq 0$. The tuple $B(u) \in \mathrm{FD}(O)$ has at least $d(O^\perp)$ nonzero blocks. At most one is the target block, so every lift has at least $d(O^\perp)-1 > r$ helpers outside it. Thus no nonzero-functional sector in (9) occurs at cost at most r.

The zero-functional sector has exact cost $\mu_{I,P,T}(0)+d(I^\perp) = \rho_T(I)+d(I^\perp)$. Theorem 4.1 now gives both directions of (22) and the support-preserving bijection. ☐

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 4.11 (Confinement by recovered dimension under an outer-distance condition)**.** *Let r be a nonnegative integer, let $O \leq L^N$ be L-linear with $N \geq 2$ and $d(O^\perp) > r+1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost-$\leq r$ recovery system for every helper-recoverable t-dimensional target subspace if and only if*

$$r < M_t(D_P,K_P) + d(I^\perp). \tag{23}$$

When (23) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r.

Proof. By Theorem 3.3, every such target subspace has inner cost at least $M_t(D_P,K_P)$, and some target subspace attains this minimum. Apply Theorem 4.10 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. ☐

4.5 One-coordinate specialization

For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \mathrm{wt}(y), \qquad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \mathrm{wt}(y),$$

with value ∞ when a constrained fibre is empty. The first quantity is the usual coset-leader weight for the inner encoder, while the second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [8, Theorem 2.1]. Here a dual word "through x" simply means that its coefficient at x is nonzero. The quantity below counts total dual-word weight; the general recovery cost excludes the forced target coordinate, so for a nonzero target column the two conventions satisfy $Z_{j,x} = \Gamma_{j,T} + 1$.

Theorem 4.12 (Exact weighted finite-length confinement at one coordinate)**.** *Fix an outer block j, an inner coordinate x, and $N \geq 2$, and assume that the coordinate projection $O \to L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j,x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O,I) = \min\left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h)\in \mathrm{FD}(O) \\ \beta\neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h\neq j} \lambda_I(\beta_h) \right) \right\}. \tag{24}$$

A minimum over an empty set is ∞. Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j,x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O,I) > r + 1. \tag{25}$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks and apply (5). For a fixed nonzero functional tuple, the block representatives minimize independently, giving the second term of (24). For the zero tuple, nonconfinement costs $\mu_{I,x}(0)$ in the target block and $d(I^\perp)$ in another block. The projection hypothesis makes confined words exactly the zero-extensions of inner-dual words, so these two sectors prove (24).

If the target column is nonzero, take $P = \{x\}$ and let $T \leq U_P$ be its span. After choosing a nonzero vector of T, a linear map $T \to L^*$ is one functional. Normalizing its target coefficient and rescaling the functional tuple identify the minima in (9) with those in (24). The target coordinate contributes one to total word weight and is not a helper, so

$$Z_{j,x}(O,I) = \Gamma_{j,T}(O,I) + 1.$$

Theorem 4.1 gives the claimed condition. If the target column is zero, the same condition follows directly from the exact minimum already proved. In either case, below the minimum every equation is confined, so restriction and zero-extension are inverse. In one dimension every exact helper support is the support of its normalized equation; the equivalence also holds for exact helper-support families and their inclusion-minimal members. $\square$

Theorem 4.12 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code. Its construction chooses an outer dual repetition line whose projective multipliers avoid every ratio between low-cost inner labels. Consequently the labelled criterion holds even though ordinary distance sees no margin.

Proposition 4.13 (A family from a Singer cycle without the outer-distance condition)**.** *Let $k \geq 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k+1)^2. \tag{26}$$

Let I be the $[k+1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k+1 \leq n \leq 2k+1$, there is an L-linear length-n outer code O with $d(O^\perp) = n$ such that (25) holds at helper radius $n-1$ for every inner coordinate and every outer block. Thus all normalized radius-$(n-1)$ recovery equations and their exact helper supports are copied blockwise although the usual outer-distance condition fails: here $d(O^\perp) = n$ is not strictly greater than $r + 1 = n$.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k+1$ coordinate functionals of the encoder give a set S of $k+1$ points in the projective space on L^*. The multiplication maps of L, modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k-1)/(q-1)$, acting regularly on its points [43]. Here regular means that a unique cycle element sends any chosen point to any other. Each ordered pair of points of S therefore forbids at most one cycle element, so at most $|S|^2 = (k+1)^2$ elements send a point of S to a point of S. Thus (26) permits a multiplication map $a : L \to L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^\times$, whose dual action satisfies $a^*(S) \cap S = \varnothing$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \ldots, f)$. If $f \neq 0$, all n entries are nonzero, so $d(O^\perp) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k+1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S, contrary to the choice of a. Every nonzero tuple therefore costs at least $n+1$.

Every outer coordinate projection is surjective. Finally, $I^\perp$ is spanned by $(1, \ldots, 1, -1)$, so $d(I^\perp) = \mu_{I,x}(0) = k+1$ for every x. The zero-functional cost in (24) is $2k+2 > n$, and the nonzero-functional cost is at least $n+1$. Thus $Z_{j,x}(O, I) > n$ for every j, x. At $r = n-1$ the ordinary distance condition would require $d(O^\perp) > n$, which is false, while the exact weighted condition holds. Taking $(q, k, n) = (5, 3, 5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n - 1 \geq k$, the asserted transfer is nonvacuous throughout the family. □

For a nondegenerate singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x. If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp). \tag{27}$$

The helper-radius condition (23) is therefore $r + 1 < z_x(I)$, with no change of convention hidden in the reduction. Theorem 4.1 is the common finite formula. The outer-distance condition removes its nonzero-functional sector and gives the RGHW threshold in every recovered dimension. Restricting instead to one target coordinate gives Theorem 4.12 without that condition.

The labelled functions are sufficient for exact numerical composition; minimizing lifts must also be stored to propagate witnesses. Their coarsenings answer different questions. The next section shows sharply that relative-weight minima do not determine recovery reliability and that the abstract nested pair does not determine confinement.

4.6 Relation to adjacent formulations

RGHWs and RDLPs describe support and access thresholds for nested codes [29, 25, 16, 41, 15]; our identity $\mu_t(P) = M_t(D_P, K_P)$ specializes this theory to the canonical target/helper pair. Recovery-set formulations retain supports but generally discard the normalized coefficients and functional labels coupling blocks [17, 37, 40, 1, 32, 18]. Disjoint simplex recovery likewise optimizes support families rather than labelled transfer [7]. Regenerating codes instead optimize download and subpacketization [42, 45, 20, 33].

Service-rate regions allocate requests among recovery sets under capacities [3, 4]; we transfer only the radius-bounded region generated by preserved supports. Generalized coset weights minimize supports of prescribed-coset representatives, while generalized covering radii maximize the analogous quantity over coset tuples [46, 10]. Our finite objective also retains target normalization and the functional labels imposed by the outer code.

Concatenation and matrix-product constructions produce LRCs with prescribed parameters [27, 14, 22]; Jin and Fu, for example, fix a binary $[3, 2, 2]$ inner code and choose $\mathbb{F}_4$ outer codes. We allow an arbitrary inner code and target split and identify the exact finite obstruction to preserving every bounded dual recovery equation. Outer distance is needed only to collapse that obstruction to the RGHW formula. We make no minimum-bandwidth claim.

The functional-dual decomposition is classical, and min–sum elimination also appears in concatenated decoding [13, 8, 2, 19, 5]. Symbol-indexed soft information, presentation-dependent input–output enumerators, and message weights retain labels for decoding or enumeration [28, 36]; our target-normalized costs instead identify the first nonconfined support.

Contextual equivalence and congruence are established ideas in automata and finite-interface decomposition. Weighted-automata minimization identifies states by their future weighted behaviour [35], and analogous Myhill–Nerode statements exist for recognizable tree series [31]. In coding and represented-matroid settings, minimal tree realizations and finite equivalence across bounded-width separations likewise compress interface information [23, 12]. We import none of those theorems. Here the contexts are outer linear codes, the response is the exact target-normalized recovery-support cost, and Theorem 4.7 proves a radius-controlled separator bound and bounded witness cover. The witness-propagating min–sum composition is Theorem 4.4.

5 Sharp limits of coarser recovery invariants

The preceding sections identify exact information for three different tasks. Here we show that two tempting coarsenings lose essential information. The first separation rules out the complete RGHW hierarchy as a reliability invariant at every quotient rank: minimum support sizes do not retain the overlap pattern of recovery sets. The second rules out even the full abstract nested pair, together with the same unlabelled helper supports, as a confinement invariant: its ambient presentation can change $d(I^{\perp})$ and the prescribed-coset costs in (9).

First consider a one-dimensional recovered space. On the helper set $\{0, \ldots, 8\}$, let

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\},$$

where, for example, 158 denotes $\{1, 5, 8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10, 4, 6]_q$ code over a common finite prime field. Here a *clutter* means a family of sets none of which contains another; in this example it is the complete radius-three family of inclusion-minimal recovery supports. Larger minimal supports also occur: in

$\mathcal{A}$, $\{0,1,2,4\}$ contains none of the listed triples, while the construction below makes its four helper columns a basis and hence a recovery set.

Proposition 5.1 (Rank-one support-overlap separation)**.** *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s, their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \tag{28}$$

$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \tag{29}$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \ldots, 5$, the numbers of k-edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9.

Inclusion–exclusion for the five edge-containment events gives (28) and (29).

We now give the representation argument. Its purpose is to realize exactly the five listed triples as collinear triples; after lifting, adjoining the distinguished target x turns exactly those triples into four-column circuits.

The two plane arrangements. Over an infinite field K, choose five projective lines $\ell_1, \ldots, \ell_5$. For $\mathcal{A}$, make ℓ_2, ℓ_3, ℓ_4 concurrent and label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose $2, 6$ on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. The incidence skeleton is

	three concurrent lines	other prescribed intersections
$\mathcal{A}$	ℓ_2, ℓ_3, ℓ_4 at 0	$1 = \ell_1 \cap \ell_4,\ 4 = \ell_3 \cap \ell_5,\ 5 = \ell_1 \cap \ell_3,\ 8 = \ell_1 \cap \ell_5$
$\mathcal{B}$	ℓ_1, ℓ_4, ℓ_5 at 1	$8 = \ell_1 \cap \ell_3,\ 7 = \ell_2 \cap \ell_3,\ 0 = \ell_3 \cap \ell_4,\ 2 = \ell_2 \cap \ell_5$

At each free choice, avoid the finitely many unwanted joining lines. The unwanted lines are precisely those that would create an additional collinear triple. The displayed triples are therefore exactly the collinear triples.

Generic lift. Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i. For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i. For four helper points, the quotient vectors span K^3, so their lifted determinant is a nonzero linear form in the four first coordinates. Avoiding this finite union makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A.

Finite-field realization and parameters. Choose the construction over $\mathbb{Q}$ and reduce both configurations modulo one prime avoiding the finitely many nonzero determinants. The two row-span codes then have rank four, dual distance four, and minimum recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. □

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius-R full-quotient reliability by

$$\Pr\{\exists L \leq D :\ L + K = D,\ |\operatorname{supp} L| \leq R,\ \operatorname{supp} L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s. The condition $L + K = D$ says that the surviving helper equations recover the entire quotient D/K; the radius R bounds their union support $|\operatorname{supp} L|$.

Theorem 5.2 (The complete RGHW hierarchy does not determine reliability)**.** *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \ldots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 5.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \qquad (1 \leq i \leq \ell - 1),$$

where each padding line has the unique support of size L_i. Thus

$$K = K_0 \oplus 0 \oplus \cdots \oplus 0, \qquad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \cdots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the tth relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \ldots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t-dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R. Thus

$$R_{\text{full},R}(s) = s^{\sum_i L_i} R_{\text{base},3}(s).$$

The two polynomials remain different by (28)–(29).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J / K$, let $\pi : \mathbb{F}_q^J \to V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \to D/K \leq V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \qquad G(a, x) = \alpha(a) + \pi(x),$$

uses the first ℓ coordinates as targets, J as helpers, and has message space V. Its helper kernel is K and the helper preimage of the target span is D, so its associated nested pair is $K \subseteq D$. □

The second separation fixes the abstract nested pair itself. This is not a row-basis change of one inner code: by Proposition 3.2, such a change leaves (D_P, K_P) unchanged. Here a *coefficient presentation* means an ambient inner dual code, target coordinates, and target coefficient map that realize a prescribed abstract pair $K \subseteq D$ as its shortening–puncturing pair. Different such realizations may have different ambient dual codes and distances, so the examples below cannot be related by changing the row basis of one encoder. Let

$$K = 0, \qquad D = \langle u, v \rangle \leq \mathbb{F}_2^3, \qquad u = 100, \quad v = 011.$$

The nonzero helper words have weights $1, 2, 3$, so $(M_1, M_2) = (1, 3)$.

Proposition 5.3 (The abstract nested pair does not determine confinement)**.** *Present the target space first by*

$$10 \mapsto u, \qquad 01 \mapsto v, \qquad 11 \mapsto u + v,$$

and then by

$$11 \mapsto u, \qquad 10 \mapsto v, \qquad 01 \mapsto u + v.$$

The two inner codes have the same associated nested pair, the same unlabelled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2, 3, 5\}$ *and* $\{3, 3, 4\}$*. In every outer-distance regime where the collapse theorem applies, their additive costs are respectively*

$$(M_1 + d(I^\perp), M_2 + d(I^\perp)) = (3, 5) \qquad \textit{and} \qquad (M_1 + d(I^\perp), M_2 + d(I^\perp)) = (4, 6).$$

Proof. Use the graph-code realization

$$I^\perp = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification from the two-dimensional target space to D. Over $\mathbb{F}_2$ the minus sign is invisible. The three nonzero graph words are

	target coefficient	helper word	target/helper weights	total weight
I_1	10	100	$1 + 1$	2
	01	011	$1 + 2$	3
	11	111	$2 + 3$	5
I_2	11	100	$2 + 1$	3
	10	011	$1 + 2$	3
	01	111	$1 + 3$	4

Thus $d(I^\perp)$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1, 3)$ gives the two threshold sequences by Theorem 4.11. □

These are limits on representation, not on executability. Keeping the functional labels avoids the second loss, and keeping minimizing lifts permits the exact recursion to return the original coefficient systems. The next section turns that closure law into an optimizer.

6 Exact recovery optimization

The closure law in Theorem 4.4 supplies an exact compiler state: its coordinates are the intermediate functionals that scalar summaries discard. ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) has developed into a compiler and exact solver for finite algebraic optimization, with recovery as one application. Its standalone repository is `tavisrudd/ergodis`; the software is developed and distributed separately from this manuscript. Numerical tables store fibrewise minima; predecessor labels and minimizing lifts reconstruct coefficient-level witnesses. Throughout this section, the algebraic cost counts participating scalar helper coordinates. Repair bandwidth, subpacketization, and physical data movement are separate downstream objectives.

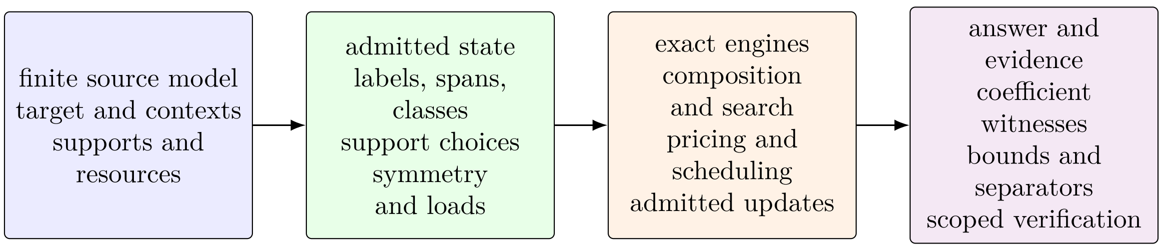

Figure 2: Ergodis separates the source model, admitted representation, optimization, and evidence contract. Functional labels preserve composition; support alternatives preserve overlap and helper-price queries. Stored lifts reconstruct feasible decisions, while optimality needs a matching checked bound or complete search justification. Reuse under updates or new readouts requires admission for the chosen observation. The mathematical results do not depend on executing this pipeline.

Implemented interfaces. The development system examined in September 2026 includes the following layers. Their contracts are documented in the software's optimization guide, architecture document, and interface-specific documentation; release snapshots can contain a smaller surface.

- Labelled composition and generated-span kernels support scalar and simultaneous recovery, compressed support families, and capacity-aware scheduling. Pareto load search is complemented by Lagrangian bounds and branch-and-bound; an exact primal assignment and a matching rounded dual bound certify the number of repaired demands when that bound closes.
- Finite typed deterministic presentations admit observational minimization and replayable distinguishing contexts. A separate readout admission operation checks whether a new observation is constant on each retained class, returning a concrete conflicting pair when it is not. This explicit-carrier path does not establish a construction bound for implicit presentations.
- Coordinate-orbit covers, invariant linear blocks, and connected-support CSS-distance search expose algebraic structure before residual search. Orbit coverage still requires the application to establish invariance of feasibility and of the current query's objective. Changed failures or prices can reduce the admissible symmetry group.
- Portable campaign control, native and browser execution surfaces, and saved-run records support proposing, checking, executing, inspecting, and forking bounded investigations. Candidate generation and persistence are separate from mathematical admission and the solver's hot loops.

Verification contracts. The independent `ergodis-verify` crate has no optimizer dependency. Its bounded binary-composition checker establishes a specified necessary-coordinate restriction by direct finite checking. Its separate min-plus transition checker authenticates supplied four-by-four summaries against a retained tree, with linear storage and logarithmic update work. The latter uses explicitly bounded saturating arithmetic. It certifies summary composition and replacements, not the lowering of a source model or domain optimality. Saved evidence must be checked under its declared contract; storing or reopening a record supplies no mathematical authority. These executable checkers are distinct from the paper's Lean formalization.

The theory below isolates the recovery interfaces underlying this broader system. In particular, witness reconstruction, numerical quotient minimization, preservation under all prices, and event-correct reliability are different requirements. A capability of one engine does not establish all four for every adapter.

We first state the algorithm in flattened $\mathbb{F}_q$-linear coordinates. Let T have dimension t, let the outer message space have dimension m, and let the inner label space have dimension s. For the ith outer coordinate, write

$$A_i : \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^s) \longrightarrow \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$$

for its contribution to the outer restriction map, and let $\lambda_i(b)$ be the prescribed-coset support cost of the inner label b. Different inner blocks may have different cost functions. For chosen block labels $b_1, \ldots, b_N$, the total outer label is $\sum_i A_i(b_i)$; the dynamic program enforces that this sum equals the requested label c.

Proposition 6.1 (Exact hierarchical recovery optimizer)**.** *For $c \in \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$, define*

$$D_0(c) = \begin{cases} 0, & c = 0, \\ +\infty, & c \neq 0, \end{cases} \qquad D_i(c) = \min_b \bigl(D_{i-1}(c - A_i b) + \lambda_i(b)\bigr). \tag{30}$$

Then $D_N(c)$ is the exact prescribed-coset support cost of the concatenated code at label c. Storing a minimizing label b and predecessor state at each finite entry returns an exact block-labelled recovery witness, not only its scalar cost. The same recursion applies to the target-normalized formula after adjoining its target-image labels to the state.

If every label is enumerated, put $Q = q^{mt}$ and $S = q^{st}$. The direct table algorithm uses $O(NQS)$ min–sum transitions, $O(Q)$ working cost memory, and $O(NQ)$ predecessor entries when all witnesses are retained.

Proof. After i blocks, $D_i(c)$ is the least sum of inner support costs among labels $(b_1, \ldots, b_i)$ satisfying $\sum_{j \leq i} A_j b_j = c$. This invariant is true at $i = 0$. Conditioning on the last label b_i gives exactly (30), so induction proves the claim. The leaf supports lie in disjoint blocks, hence their costs add. Backtracking the stored minimizers reconstructs the labels; the inner prescribed-coset witnesses then reconstruct the coefficient-level recovery system. There are Q states and S candidate labels per block, which gives the stated bounds. The target-normalized case is the same invariant applied to the paired labels in (13). $\square$

The implementation also removes unreachable labels and merges states only when an explicit expansion map identifies each reduced choice with original block labels of the same cost. Backtracking through that map recovers an original-instance witness. Quotient coordinates, projectively equivalent columns, and repeated block types are instances of this implementation rule; they are not additional claims of Proposition 6.1. Proposition 4.7 supplies a separate theorem-backed reduction for composition: every target-rank-t witness factors through the L-span generated by its image, whose dimension is at most t. An exact search may therefore cache that generated span and discard states of larger dimension without losing an optimum or witness.

The resulting table supplies all exact helper costs needed by the finite confinement formula, including its zero and nonzero outer-functional sectors. It can therefore emit a replayable recovery witness when a threshold fails, or a complete exhausted-state certificate when the requested label is unreachable. Iterating the compiler through a tower preserves the intermediate labels required at the next level; collapsing each table to one minimum would destroy precisely this compositional information.

6.1 Interfaces for resource choices and availability

To reuse compilation under changing prices or failures, retain more than a unit-cost minimizer. For each ordinary label b, let

$$\mathcal{A}_{i,T}(b) = \min_{\subseteq}\{\operatorname{supp} Y(T) : \Phi_i Y = b\},$$

where $\min_{\subseteq}$ removes strict supersets and the empty fibre gives the empty family. The family $\{\varnothing\}$ is distinct from the empty family. In the notation of the flattened optimizer, the exact substitution is

$$\mathcal{A}_T^{\mathrm{out}}(c) = \min_{\subseteq}\left\{\bigcup_i H_i : \sum_i A_i b_i = c, \quad H_i \in \mathcal{A}_{i,T}(b_i)\right\}. \tag{31}$$

Indeed every lift decomposes into these block fibres; replacing one block support by a contained support with the same label preserves compatibility. Conversely every displayed choice lifts. The same argument applies to Theorem 3.5, and to target normalization with its existing constraints. Retain a coefficient witness for each surviving support. For additive nonnegative resource loads one may instead retain a Pareto frontier within each label: coordinatewise domination is preserved by addition. This is safe only for the resource observations represented by that load vector. The numerical contextual quotient in Corollary 4.8 does not automatically preserve these richer observations.

Proposition 6.2 (Supports, prices, and availability)**.** *Fix a finite labelled helper set E, a radius, and its minimal recovery family $\mathcal{M}$. The following data determine one another: $\mathcal{M}$; the success indicator $F(A) = \mathbf{1}\{\exists H \in \mathcal{M} : H \subseteq A\}$ on every $A \subseteq E$; the price function*

$$\pi_{\mathcal{M}}(p) = \min_{H\in\mathcal{M}} \sum_{h\in H} p_h \quad (p \in \mathbb{R}^E_{\geq 0});$$

and the reliability for every independent helper survival vector in $[0,1]^E$. Empty price minima have value ∞.

Proof. The family gives both displayed functions and its reliability by summing $F(A)$ against the probability of each available set. The minimal successful sets of F recover $\mathcal{M}$. Prices zero on A and one outside A satisfy $\pi_{\mathcal{M}}(p) = 0$ exactly when $F(A) = 1$, including the empty-family case. Finally deterministic survival vectors recover every $F(A)$ from reliability. These implications prove the assertion. □

This equivalence concerns information, not the number of oracle calls. Extracting reliability can require exponentially many availability queries. Summing probabilities of coefficient witnesses is not event evaluation: repairs $\{1,2\}$ and $\{2,3\}$ with survival probability $1/2$ each give success probability $1/4 + 1/4 - 1/8 = 3/8$, although summing the two repair probabilities gives $1/2$. One must evaluate the existential event F, for example by inclusion–exclusion or a Boolean representation that retains overlap. Factoring probabilities is valid on independent variable sets; adding alternative probabilities is valid on disjoint events.

For service optimization, retain the radius as a separate constraint and evaluate $\pi_a(p)$ over supports of size at most r. An alternative discarded under unit prices can become optimal under new prices. Either the support families or a valuation-independent circuit of feasible choices permits exact repricing and reconstruction. For the fractional objective $\max \sum_a w_a \lambda_a$ under helper capacities c_h, the full dual is

$$\min_{p \geq 0} \sum_h c_h p_h, \qquad \sum_{h \in R} p_h \geq w_a \quad (a \in \mathcal{A},\ R \in \mathcal{M}_a^{(\leq r)}).$$

Thus solving a restricted service LP, pricing every demand, and adding a witness whenever $w_a - \pi_a(p) > 0$ yields column generation. If the restricted primal and dual have equal finite values and no price violation remains, their value is the full optimum: the restricted primal is feasible for the full problem and the prices give a matching full dual bound. This is a fractional certificate. Integral scheduling needs its own exact search or bounds. Support-family equality transfers an integral feasible region, but an LP optimum need not be an integral optimum.

6.2 Width of a fixed query

The ambient syndrome dimension is a pessimistic state bound. Fix c and write $A_i b = a_i \circ b$, where $a_i : L_i \to W$ is linear and $b : T \to L_i$. On a supplied binary tree of factors, a subtree S has contribution spaces

$$U_S = \sum_{i \in S} \operatorname{im} a_i, \qquad V_S = \sum_{i \notin S} \operatorname{im} a_i, \qquad w_S = \dim(U_S \cap V_S).$$

Proposition 6.3 (Boundary-coordinate compilation). *For a fixed linear output map $c : T \to W$, $\dim T = t$, feasible partial syndromes at cut S lie in*

$$\operatorname{Hom}(T, U_S) \cap \big(c - \operatorname{Hom}(T, V_S)\big).$$

This set is empty or an affine space with q^{tw_S} elements. Given local choice tables on n disjoint blocks and a binary tree with $w = \max_S w_S$, scalar additive optimization uses at most $O(nq^{2tw})$ pair combinations, in addition to scanning the local tables and polynomial linear algebra. It returns an attaining lift when local lifts are supplied. A separate integer support budget r increases the bound to $O(n(r+1)^2 q^{2tw})$ combinations.

Proof. Subtract any feasible partial syndrome. The difference space is $\operatorname{Hom}(T, U_S) \cap \operatorname{Hom}(T, V_S) = \operatorname{Hom}(T, U_S \cap V_S)$, of dimension tw_S. Choose affine boundary coordinates by linear algebra. At a leaf scan its table and retain the minimum at each admitted boundary state. At an internal node enumerate the pairs of child states, add their syndrome maps, discard sums outside the parent affine space, and minimize their summed cost at each parent state. Induction on the tree proves the same invariant as in Proposition 6.1. Each child has at most q^{tw} states, giving the pair bound. Store minimizers and backtrack for witnesses. For a support budget retain the best price at each exact helper count $0, \ldots, r$ and add counts when combining; disjoint ownership makes those counts additive. ☐

The bound charges local compilation separately. It counts arithmetic operations; exact rational prices also incur their arithmetic bit cost. Paired normalization labels must be included when measuring width for a target-normalized query. Support or Pareto messages multiply combination work by the numbers of retained alternatives and their pruning costs. On a balanced tree, a change to one leaf's costs or availability, with the linear interfaces and fixed query unchanged, recomputes only its ancestor messages: $O(C_{\text{combine}} \log n)$ work plus the leaf update. Changing the interface spaces can require new coordinates. None of these bounds avoids the output cost of explicitly requesting all $q^{t \dim W}$ syndromes.

An independent optimality check can use the layered transition graph of the fixed-query computation. A feasible path of cost C, together with potentials satisfying $p(v) \le p(u) + w(u, v)$ on every legal edge, $p(\text{source}) = 0$, and $p(\text{terminal}) = C$, proves optimality by telescoping along every source–terminal path. This requires complete transition coverage and justified local costs and state reductions; a reconstructed path alone supplies only an upper bound.

6.3 A worked bilinear task reduction

Let worker i return $a_i(x)b_i(y)$, with $a_i \in V^*$ and $b_i \in W^*$ over $\mathbb{F}_q$. For $\tau_i = a_i \otimes b_i$, define $Q : \mathbb{F}_q^E \to V^* \otimes W^*$ by $Qe_i = \tau_i$. A bilinear target f has a linear aggregation on workers H exactly when $f = \sum_{i \in H} c_i \tau_i$: evaluate on pairs of basis vectors to prove necessity, and use bilinearity for sufficiency. Taking $D = Q^{-1}(\langle f \rangle)$ and $K = \ker Q$ makes this a prescribed quotient lift when $f \in \operatorname{im} Q$.

For example, three workers return

$$x_1 y_1, \qquad x_2 y_2, \qquad (x_1 + x_2)(y_1 + y_2),$$

and the target is $x_1 y_2 + x_2 y_1$. Comparing the four tensor coefficients forces the aggregation coefficients $(-1, -1, 1)$ over every field, including characteristic two. All three workers are necessary, the contacted-worker cost is three, and the nonnegative weighted cost is $p_1 + p_2 + p_3$. If any worker is unavailable the target is outside the available span. Generally, a separating linear functional that vanishes on that span but not on f certifies this impossibility, by extending a basis of the span. It certifies impossibility of the stated linear aggregation protocol.

Compression to a smaller worker-function space is exact when a linear map is injective on the span of the workers and requested tasks: every coefficient relation then holds before compression exactly when it holds afterward. This preserves supports, prices, and aggregation witnesses. A Schur-product presentation can be used when it supplies this identification. For repeated inputs or polynomial tasks, the space must be the actual function space; formal monomials need not be independent over a finite field.

6.4 Measured performance and solver choice

The measurements below are the previously recorded recovery benchmark snapshot, not a measurement of every current engine. They assess engineering reach and the point at which a general solver becomes preferable; they are not mathematical evidence or the principal contribution of the software. The structural comparison is how interface width, target rank, field size, distinct block types, frontier size, and query reuse affect total cost. Compilation, repeated queries, witness expansion, and verification must be charged separately; a fast retained-state query does not imply a fast construction of that state.

The specialized dynamic programs are most useful when compilation exposes a small algebraic state space. They are not replacements for a general constraint solver when the residual model contains unrelated side constraints. The appropriate deployment rule is therefore:

$$\text{algebraic compilation} \longrightarrow \begin{cases} \text{specialized exact DP}, & \text{when the compiled state is sufficient,} \\ \text{CP-SAT}, & \text{for the remaining general constraints.} \end{cases}$$

CP-SAT combines constraint programming, satisfiability, and linear-relaxation methods [39]; it is therefore an appropriate end-to-end baseline. A second, *structured CP-SAT* baseline receives the same symmetry reduction and algebraic preprocessing as ergodis, isolating the value of the specialized state engine from the value of compilation itself.

Measurement protocol. The measurements were made on an AMD Ryzen AI 9 HX 370 under Linux. Each paired program used the same selected logical CPU, and their order was rotated to limit ambient-load bias. Elapsed times are monotonic in-process times and include construction of the compiled state or control model. The application controls use eleven rounds and the Rust kernels seven; reported values are medians of the raw samples. The only exceptions are explicitly identified single completed deterministic solves. Peak memory is the process high-water resident set size. Exact versions, checksums, raw samples, and replay commands are recorded in `evidence/benchmarks.json` of the ergodis repository (`https://github.com/tavisrudd/ergodis`).

The cloud-LRC scheduling kernel in Table 2 operates on the exact six-type quotient of the 100,000 demands and is timed in a repeated inner loop; its displayed "$< 1\,\mu$s" value is therefore an amortized solve time for the preaggregated six-type instance, not a claim about parsing 100,000 input records. We use only the conservative ratio implied by that displayed upper bound.

Compiler versus residual engine. The benchmark suite separates the algebraic compiler from the residual state engine. On four exact capacitated-scheduling profiles, ergodis was 2.5–377 times faster than direct CP-SAT and 2.5–373 times faster than structured CP-SAT receiving the same safe preprocessing. The represented tower benchmark isolates the composition theorem: at depth six and fanout four, ergodis expands an exact coefficient witness over 4096 leaves in a 4.559 ms cold-process median. Direct CP-SAT exceeded 400 s in each of three paired rounds, giving a speedup greater than 87,743; for equal eight-solve batches the bound is greater than 45,103. The labelled-table control has not been rerun under this corrected protocol.

The same compilation pattern has been tested through six bundled application examples. These are separate matched comparisons, not a ranking of unlike tasks. Each row below states the question and the exact output checked against its formulation-specific open-source control. The cold-process gaps range from 34-fold to 190-fold. The comparisons do not include commercial solvers or claim a universal ranking.

example	question/output checked	bounded instance	exact control	cold faster	warm faster
recursive XOR	complete minimal-support family	8 diamonds, 256 supports	Graphillion ZDD [21]	33.71×	22.94×
cloud LRC	maximum served count and loads	100,000 demands, six types	HiGHS integer model	160.45×	160.21×
repair DAG	optimal makespan and schedule	3×21 tasks	CP-SAT intervals	167.12×	275.15×
QC-LDPC	absence of a weight-four word	lift 50,000	CryptoMiniSat XOR	190.45×	39.72×
vector repair	minimum nodes and spanning set	64 nodes, 2 symbols/node	CryptoMiniSat XOR	75.72×	64.90×
GPU MDS	feasible flow and helper assignment	10,000 shards, 64 failures	OR-Tools max-flow	102.42×	103.23×

Table 2: Six separate exact comparisons under the corrected process-symmetric protocol. Cold runs perform one solve per fresh process; warm runs perform eight solves on both sides and normalize external wall time per solve. ZDD means zero-suppressed decision diagram; LRC means locally repairable code; DAG means directed acyclic graph; QC-LDPC means quasi-cyclic low-density parity-check code; MDS means maximum-distance-separable code. Ratios use unrounded paired samples.

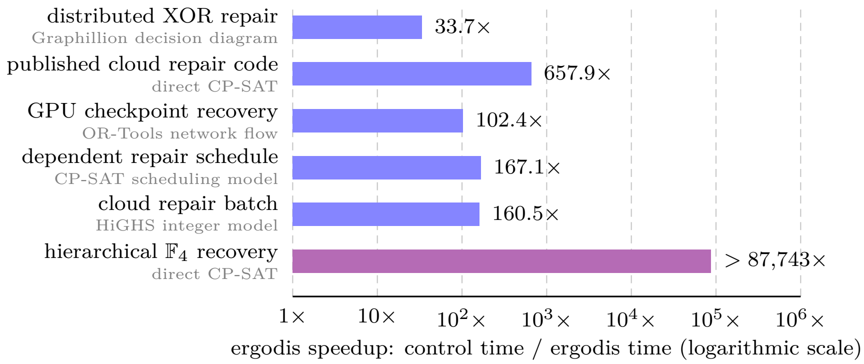

Figure 3: How many times faster ergodis is than matched exact controls for representative applications and the theorem-driven tower computation under the corrected cold fresh-process protocol. Longer bars mean larger ergodis speedups. Each formulation-specific control is printed beneath its workload; full instances appear in Table 2, with the published cloud-code check below. The cloud-code row is Jin and Fu's family [22]. The distinguished hierarchical $\mathbb{F}_4$ row compares labelled min–sum composition with direct CP-SAT and isolates the reduction supplied by the closure theorem; its bar is a 400-second timeout lower bound. Controls differ by row, so this is not a general solver ranking.

As a code-native check, ergodis also analyzes Jin and Fu's published $\mathbb{F}_4$ Hamming-outer family [22, Corollary 5.4]. For its binary $[4095, 2718, 6; 2]$ LRC, all backends obtain the same exact confinement cost. Three corrected cold-process pairs give a 442.573 ms ergodis median and a 270.939 s direct-CP-SAT median. The paired speedup is 657.88 with log-ratio $t = 49.09$. Each equal eight-solve direct-CP-SAT batch exceeded 400 s, giving a separate warm speedup greater than 196.50. The labelled-table model has not been rerun under this protocol. Exact inputs, outputs, tool versions, peak memory, replay commands, and raw timing samples are in the corrected application evidence under `evidence/` and in `BENCHMARKS.md` of the ergodis repository.

General-purpose constraint solvers have also been used to construct binary codes [24], and repair dependency graphs and scheduling predate this work [26]. The contribution here is the structure-aware compilation of linear-recovery fibres, with exact witnesses, before a specialized dynamic program or a residual solver is invoked.

The optimizer changes no mathematical hypothesis: it evaluates the labelled state proved sufficient above. We now return to consequences of the collapsed RGHW threshold, beginning with ordinary generalized weights and extremal inner codes.

7 Consequences of the threshold

The fixed-target cost refines ordinary generalized weights, which optimize over target sets, and is bounded above by worst-case cooperative locality, which maximizes over them. Let $d_e(C^\perp)$ denote the eth generalized Hamming weight of $C^\perp$ [44]. For an e-set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e-dimensional subcode $A \leq C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$, so every target coefficient vector has a unique equation in A. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

The first identity below is the standard information-set interpretation of generalized Hamming weights; we record it to derive the paper-specific nonconfinement consequence.

Theorem 7.1 (Best-target generalized weight and first nonconfinement cost)**.** *For* $1 \le e \le \dim C^{\perp}$*,*

$$\min_{\substack{P\subseteq E\\ |P|=e}} \kappa_C(P) = d_e(C^{\perp}) - e. \tag{32}$$

Consequently, if every e*-set is recoverable and* $r_e^{\mathrm{coop}}(C) = \max_{|P|=e} \kappa_C(P)$*, then*

$$r_e^{\mathrm{coop}}(C) \ge d_e(C^{\perp}) - e.$$

For a fixed inner code C*, once the outer dual distance is large enough for the collapse in Theorem 4.10, the least helper cost of a first nonconfined system over all* e*-coordinate target sets is*

$$d_e(C^{\perp}) - e + d(C^{\perp}). \tag{33}$$

Proof. A normalized identity system on P spans an e-dimensional subcode of $C^{\perp}$ whose support is P together with its helper union. Hence $d_e(C^{\perp}) \le e + \kappa_C(P)$.

Conversely, let $A \le C^{\perp}$ have dimension e and support size $d_e(C^{\perp})$. Choose e linearly independent columns of a generator matrix of A; their coordinate set P is an information set and has size e, and restriction $A \to \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\operatorname{supp}(A) \setminus P$. This proves (32); the cooperative-locality inequality follows by taking the maximum instead of the minimum; compare [1, Theorem 1].

It remains to justify the nonconfinement formula when the target columns indexed by P are dependent. Put $U = \operatorname{im} G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \to \mathbb{F}_q^P$ gives a system on U. Conversely, given a system on U, write

$$a = \alpha(G_P a) + \big(a - \alpha(G_P a)\big) \qquad (a \in \mathbb{F}_q^P).$$

Thus $G_P : \mathbb{F}_q^P \twoheadrightarrow U$ separates each target vector into its chosen lift $\alpha(G_P a)$ and a kernel relation. The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

Now repeat the block-functional argument of Theorem 4.10. Once the outer-functional case is excluded, the target block of any identity system uses at least $\kappa_C(P)$ helpers. If the system is nonconfined, a nonzero equation map in another block has union support at least $d(C^{\perp})$; the two supports are disjoint. Conversely, attach a minimum-weight word of $C^{\perp}$ in a second block through any nonzero linear functional on $\mathbb{F}_q^P$. Hence the first nonconfinement cost for this P is $\kappa_C(P) + d(C^{\perp})$. Minimizing over P and using (32) proves (33). □

Before bounding the MDS thresholds, we record what the coefficient layer adds even when the support clutter is completely uniform.

Theorem 7.2 (Minimum-weight MDS recovery equations span the dual)**.** *Let* $C \le \mathbb{F}_q^E$ *be an* $[n,k]_q$ *MDS code with* $1 \le k < n$*, and let* $x \in E$*. Then*

$$\operatorname{span} \mathcal{P}_x^{(\le k)}(C) = C^{\perp},$$

and their supports form the complete k*-uniform clutter on* $E \setminus \{x\}$*.*

Equivalently, the minimum-weight dual equations through x realize every k-subset of the other coordinates as a helper support.

Proof. The dual has dimension $n-k$ and distance $k+1$. For each $(k+1)$-set T, restriction $C^\perp \to \mathbb{F}_q^{E\setminus T}$ has a nonzero kernel, whose words have support exactly T by the distance bound. Thus all stated supports occur. Fix a $(k-1)$-set $Q \subseteq E \setminus \{x\}$ and take the normalized word on $\{x,t\} \cup Q$ for each $t \in E \setminus (\{x\} \cup Q)$. Their private coordinates t make these $n-k$ words independent, hence a basis of $C^\perp$. □

The support clutter depends only on the MDS parameters, whereas the normalized equations retain the represented dual code.

Put $k = \dim I$, $u = \operatorname{rank} G_P$, and $b = \operatorname{rank} G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \le b - \ell + t, \qquad d(I^\perp) \le k+1. \tag{34}$$

The three ranks governing the statement are

$$\begin{array}{r|l} u = \operatorname{rank} G_P & \text{dimension visible on the target coordinates} \\ b = \operatorname{rank} G_J & \text{dimension visible on the helper coordinates} \\ \ell = \dim(\operatorname{im} G_P \cap \operatorname{im} G_J) & \text{dimension recoverable from the helpers.} \end{array}$$

Theorem 7.3 (MDS ceiling and rank-one rigidity)**.** Universal ceiling. *For $1 \le t \le \ell$,*

$$M_t(D_P, K_P) + d(I^\perp) \le 2k - \ell + t + 1. \tag{35}$$

MDS staircase. *Suppose I is MDS. Then*

$$u = \min\{k, |P|\}, \qquad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \ge 1$, then, for $1 \le t \le \ell$,

$$M_t(D_P, K_P) = k - u + t, \qquad M_t(D_P, K_P) + d(I^\perp) = 2k - u + t + 1. \tag{36}$$

If $|J| \ge k$, then $\ell = u$ and these values attain the ceiling in (35) at every rank. Rank-one rigidity. *Conversely, when $\ell \ge 1$, equality in (35) at $t = 1$ forces I to be MDS and forces equality at every rank.*

Proof. The ceiling follows from (34) and $b \le k$. For an MDS code every set of at most k generator columns is independent; equivalently, the column matroid is uniform. Thus $\operatorname{rank} G_P = u$, $\operatorname{rank} G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \le k$,

$$\dim\bigl(\operatorname{span} G_P \cap \operatorname{span} G_H\bigr) = \max\{0, u + h - k\}.$$

This is the usual dimension count for two subspaces generated by disjoint subsets of a uniform k-dimensional column configuration. Thus Proposition 3.4 gives $M_t = k - u + t$ for $1 \le t \le \ell$; here $k - u + t \le b \le |J|$. Now $d(I^\perp) = k+1$, and if $|J| \ge k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \le k$. Hence $d(I^\perp) = k+1$; equality in the dual Singleton bound is precisely the MDS condition for I. Also $M_1 = b - \ell + 1$. Strict growth then gives $M_t \ge b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. □

Thus extremality is detected at rank one: equality in the first threshold forces the inner code to be MDS and forces every later rank to attain its ceiling. No separate higher-rank hypothesis is needed.

We next show that the transferred local structure can occupy a positive fraction of an asymptotically good concatenated family, rather than appearing only in one distinguished block. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $\Delta_N = d(O_N)$.

Corollary 7.4 (Positive-density realization). *Assume $d(O_N^\perp) \to \infty$. For every fixed finite radius r, the inclusion-minimal recovery supports for every nonzero $T \leq W_P$ are eventually copied in every block, with no inner additive ceiling on r. For the stronger coefficient statement, fix either a nonzero target subspace T and a radius r satisfying Theorem 4.10, or a rank t and radius r satisfying Theorem 4.11. For all sufficiently large N, every corresponding inner normalized recovery system of cost at most r and its exact helper support are copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density $|P|/m$.*

In particular, if $r < M_1(D_P, K_P) + d(I^\perp)$, then all cost-$\leq r$ systems for all nonzero subspaces of W_P are copied simultaneously in every block by Corollary 4.9. Hence the complete bounded local recovery data through radius r occur on the stated positive-density coordinate classes.

The concatenated code has length mN, dimension kK_N, and minimum distance at least $d\Delta_N$. If

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{\Delta_N}{N} \geq \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families with positive limiting rate, positive primal relative distance, and dual distance tending to infinity exist over every finite field L.

Proof. Minimal-support transfer follows from Theorem 4.2, since $\delta_j(O_N^\perp) \geq d(O_N^\perp)$ uniformly in j. The coefficient transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N. A nonzero outer word has at least Δ_N nonzero symbols, each encoding to an inner word of weight at least d. This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^\perp$ such that

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R,$$

where H_Q is the standard Q-ary entropy function measuring the exponential growth rate of a Q-ary Hamming ball. The first-moment argument for a random K_N-dimensional L-linear code bounds the expected numbers of nonzero primal and dual words below the two cutoffs, up to subexponential factors, by

$$Q^{N(H_Q(\delta)+R-1)} \qquad \text{and} \qquad Q^{N(H_Q(\delta^\perp)-R)},$$

respectively. Both tend to zero by the displayed inequalities, so one may choose a sequence with rate tending to R, primal relative distance at least δ, and dual relative distance at least $\delta^\perp$; see also [30, Chapter 17]. □

Minimal-support transfer preserves the fractional load-balancing region used in service-rate problems. The target-touching distance condition is sufficient even when coefficient confinement fails.

Fix one inner target block with helper set J. Let $\mathcal{A}$ be a finite set of demands on that block, with demand a represented by a nonzero target subspace $T_a \le W_P$, and let $\mathcal{M}_a^{(\le r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector c, define

$$\Lambda_{\mathcal{A}}^{(\le r)}(c) = \left\{ \lambda \ge 0 : \begin{array}{l} \text{there are } f_{a,R} \ge 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\le r)}} f_{a,R}, \\ \displaystyle\sum_a \sum_{R \ni j} f_{a,R} \le c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [3, 4, 34, 9].

Corollary 7.5 (Transfer of bounded service-rate regions)**.** *Let $O \le L^N$ be an outer code with $N \ge 2$ whose projection onto the target block is nonzero. Suppose either*

$$\delta_j(O^\perp) > r + 1, \qquad \text{or} \qquad r < \Gamma_{j,T_a}(O, I) \quad \text{for every } a \in \mathcal{A}.$$

Transport capacities from the inner helper set to its copy in that block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality, because every minimal support is local. The same transfer holds for support-based integral allocations, obtained by requiring the variables $f_{a,R}$ to be nonnegative integers. Consequently, for an outer family with $d(O_N^\perp) \to \infty$, both equalities hold eventually at every fixed finite radius, without an inner-distance inequality.

Proof. Theorems 4.2 and 4.1 give minimal-support equality under the respective hypotheses. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. These operations preserve integrality, proving the integral assertion as well. □

The next family realizes all three information levels in closed form: its RGHWs give minimum recovery costs, its ambient dual distance gives the confinement thresholds under the outer-distance condition, and its projective rank distribution gives reliability beyond the RGHW hierarchy.

8 The projective-simplex family

This non-MDS family realizes the paper's information hierarchy in one model. Its relative weights determine the minimum costs, its ambient inner dual gives the collapsed confinement thresholds, and the full projective rank distribution determines reliability. Thus the family shows both what the RGHW hierarchy captures and what it forgets. Assume $m \ge 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\mathrm{PG}(m-1, q)$. Here $\mathrm{PG}(m-1,q)$ is the set of one-dimensional subspaces of $\mathbb{F}_q^m$, and A chooses one nonzero vector from each such subspace. Define the inner code by

$$I_m^\perp = \{(a, aA) : a \in \mathbb{F}_q^m\}. \tag{37}$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q-ary simplex code. The generalized-weight formula for S_m is classical; in fact every t-dimensional simplex subcode has the same support size [38]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 8.1 (Projective-simplex thresholds)**.** *For $1 \le t \le m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q-1}, \tag{38}$$

and

$$d(I_m^{\perp}) = q^{m-1} + 1. \tag{39}$$

Hence

$$C_t := M_t(S_m, 0) + d(I_m^{\perp}) = \frac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1. \tag{40}$$

In outer families with dual distance tending to infinity, the collapsed threshold C_t equals $\Gamma_{j,t}(O, I_m)$ once the outer dual distance reaches $C_t + 1$. Equivalently, at every fixed helper radius the corresponding confinement statement holds for all sufficiently large members of the family. The family is non-MDS for $m \ge 2$; for $m \ge 3$ the successive increments of C_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^{\perp}$ is a, and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m; this proves the asserted identification of the associated pair.

Let $U \le \mathbb{F}_q^m$ have dimension t. The common zero coordinates of the subcode UA are the projective points in $U^{\perp}$, a subspace of projective dimension $m - t - 1$. Thus

$$|\operatorname{supp}(UA)| = N - \frac{q^{m-t} - 1}{q-1} = \frac{q^m - q^{m-t}}{q-1},$$

independently of U. This proves (38). Every nonzero simplex word aA has weight q^{m-1}, while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (39). The confinement formula is Theorem 4.11. The non-MDS and increment assertions follow from the displayed parameters. □

question	retained information	exact answer
minimum rank-t helper union	RGHW hierarchy	$\dfrac{q^m - q^{m-t}}{q-1}$
eventual first nonconfined cost	RGHW plus $d(I_m^{\perp})$	$\dfrac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1$
rank-t recovery reliability	projective rank distribution	$R_t(s)$ in (42)

Table 3: The projective-simplex family separates the three questions. The first two use minimum-cost data; the stochastic law requires the distribution of ranks of failed projective sets.

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s, and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = \big(\operatorname{span} F\big)^{\perp},$$

which has dimension $m - \operatorname{rank}(F)$. Write $\begin{bmatrix} a \\ b \end{bmatrix}_q$ for the Gaussian binomial coefficient, the number of b-dimensional vector subspaces of $\mathbb{F}_q^a$. A projective r-frame here means r linearly independent projective points, or equivalently the projectivization of a vector-space basis of its r-dimensional span.

Theorem 8.2 (Projective rank reliability)**.** Exact probability. *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \mathrm{PG}(m-1,q) \\ \operatorname{rank}(F) \le m-t}} (1-s)^{|F|} s^{N-|F|}. \tag{41}$$

Closed form. *Writing $N_v = (q^v - 1)/(q-1)$, this equals*

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^{u} \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \tag{42}$$

Endpoint behavior. *Its endpoint terms are*

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \qquad (s \to 0), \tag{43}$$

$$1 - R_t(s) = B_{m,m-t+1}(1-s)^{m-t+1} + O((1-s)^{m-t+2}) \qquad (s \to 1), \tag{44}$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\operatorname{GL}(r,q)|}{r!(q-1)^r}$$

is the number of unordered projective r-frames.

The first endpoint says that rare successful recovery is dominated by the $\begin{bmatrix} m \\ t \end{bmatrix}_q$ minimum helper supports of size M_t. The second says that near-perfect availability first fails on a smallest projective frame whose rank exceeds the allowed failure span.

Proof. Recovery of t independent target combinations is possible exactly when $m - \operatorname{rank}(F) \ge t$, which gives (41). For a vector subspace $U \le \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice gives the total probability of failed sets spanning a fixed u-space U as

$$\sum_{V \le U} \mu(V,U) s^{N-N_{\dim V}}, \qquad \mu(V,U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \qquad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\begin{bmatrix} m \\ u \end{bmatrix}_q$ choices for U and $\begin{bmatrix} u \\ v \end{bmatrix}_q$ subspaces $V \le U$ of dimension v. Summing over $u \le m - t$ proves (42).

For $s \to 0$, a smallest surviving set that permits rank-t recovery is the complement of the points in an $(m-t)$-dimensional vector subspace. It has size M_t, and there are $\begin{bmatrix} m \\ m-t \end{bmatrix}_q = \begin{bmatrix} m \\ t \end{bmatrix}_q$ such subspaces. For $s \to 1$, failure first occurs when the failed points span dimension $m - t + 1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. ∎

At the same helper length, attaining first recovery cost q^{m-1} characterizes the projective-simplex column multiset up to scaling and permutation. This is the length-N, multiplicity-one specialization of Bonisoli's classification of linear equidistant codes [6].

9 Verification scope

The paper has a paper-owned Lean 4 companion built against a pinned Mathlib revision. It formalizes the construction of the associated nested code pair from abstract target and helper generator maps. In the notation of (1), the checked statements are

$$K_P \leq D_P, \qquad G_J(D_P) = W_P, \qquad G_J|_{D_P} : D_P \twoheadrightarrow W_P, \qquad \ker(G_J|_{D_P}) = K_P.$$

The corresponding declarations are

Paper statement	Lean declaration
$K_P \leq D_P$	`helper_ker_le_helperCodeForTargetSpace`
$G_J(D_P) = W_P$	`map_helperCodeForTargetSpace_eq_recoverableTargetMessageSpace`
surjectivity	`recoverableTargetMap_surjective`
restricted kernel	`mem_ker_recoverableTargetMap_iff`

The companion uses Lean 4.32.0-rc1 and the Mathlib commit

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

The axiom audit of all four reviewer-facing declarations reports only

$$\{\texttt{Classical.choice}, \texttt{Quot.sound}, \texttt{propext}\}.$$

These are routine logical and quotient principles used throughout Mathlib, not unproved assumptions about codes or recovery. No coding-theory result is introduced as a Lean axiom.

The formal coverage stops at the exact sequence. In particular, Proposition 3.2, Theorems 3.3, 4.1, 4.10, 4.11, and 4.12, as well as Theorems 4.2 and 3.5, Propositions 6.2 and 6.3, and Proposition 4.4, the strictness construction in Proposition 4.13 and the later consequences and examples, are human-proved and are not established by the paper-local Lean companion. The claim manifest records them as absent rather than treating the exact-sequence formalization as evidence for them.

The reliability polynomials in Proposition 5.1 follow from the displayed union-size table by inclusion–exclusion. No exhaustive computation is a premise of that proposition or of any theorem in the main proof chain. Machine-readable examples and replay checks may therefore be used to audit arithmetic without changing the logical dependence of the paper.

10 Conclusion

Exact recovery compilation fixes the operation before minimizing its physical realization. For one block, quotient support gives the relative generalized Hamming weights; across blocks, intermediate labels couple the local fibres. The quotient-lifting law and its recovery-normalized specialization compose those fibres with exact witness reconstruction.

Equation confinement and preservation of recovery alternatives have different thresholds. The former detects an unrelated external dual perturbation. The latter needs only to exclude short outer checks touching the target: then every minimal support is local at the prescribed radius. Reliability, fractional service, and integral support allocation inherit this stronger transfer statement. The overlap separation shows why minimum helper counts alone cannot answer them. Retaining all nonnegative price responses is semantically sufficient for availability, although efficient probability evaluation still needs to account for overlapping events.

The declared observation determines the compiler's admissible reductions. The finite numerical contextual quotient is exact for its cost responses; support antichains retain reweighting and availability, and resource frontiers retain their specified additive loads. Fixed-query boundary width bounds combination work once local tables are available. Direct construction of compact interfaces, including local compilation and verification costs, remains the relevant complexity question for implicit models. Ergodis's finite minimization, scheduling bounds, and dynamic verification interfaces make parts of this distinction executable, with separate evidence contracts.

One extension changes the local metric. In one-shot centralized linear repair, a helper storing $\mathbb{F}_q^{\alpha_h}$ contributes a map $Y_h : T \to (\mathbb{F}_q^{\alpha_h})^*$. Transmitting a basis of its image costs $\operatorname{rank} Y_h$ base-field symbols and supplies every required contribution. For disjoint physical-node ownership, replacing local support cost by $\sum_h \operatorname{rank} Y_h$ preserves the same fibre proof; when $\alpha_h = 1$ this is exactly helper-union cost. Shared nodes require actual transmission subspaces: two contributions U, V cost $\dim(U + V)$, which their separate ranks do not determine. Interactive and cooperative protocols require additional semantics. Likewise, the bilinear task reduction preserves specified worker-function relations and aggregation witnesses; extending it through compressed polynomial presentations requires proving that those relations survive the compression.

A different reduction retains a catalog $\mathcal{Q} \subseteq \mathcal{F}$ of executable witnesses satisfying

$$\min_{x\in\mathcal{F}} b_x(c) = \min_{x\in\mathcal{Q}} b_x(c) \qquad \text{for every admitted context } c.$$

Weighted representative families provide an established starting point for bounded forbidden-resource contexts [11]. This lower-envelope condition allows the replacement witness to depend on the context; it does not require each omitted witness to be equivalent to a retained one. A recovery-specific extension should preserve functional labels, coefficient lifts, and support budgets during catalog composition, then justify omitted scheduling constraints against the actual dual prices. Such a catalog contract is distinct from both the numerical contextual quotient and preservation under every nonnegative price vector. Its source-completeness certificates and probability approximation guarantees are separate questions from the exact transfer results proved here.

Programme perspective

The reconstruction question here concerns the information needed to answer specified downstream queries. Sufficiency theorems are paired with necessity boundaries: relative weights lose overlap, unlabelled costs lose functional alignment, and a numerical quotient can lose future resource observations. Ergodis gives this question an algorithmic form: compile a representation for declared contexts, reconstruct a decision, and check the justification for discarding alternatives. Different parts of the programme reconstruct objects, behaviours, or optimal decisions. This paper establishes the recovery-specific sufficiency statements and their information-loss boundaries.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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